

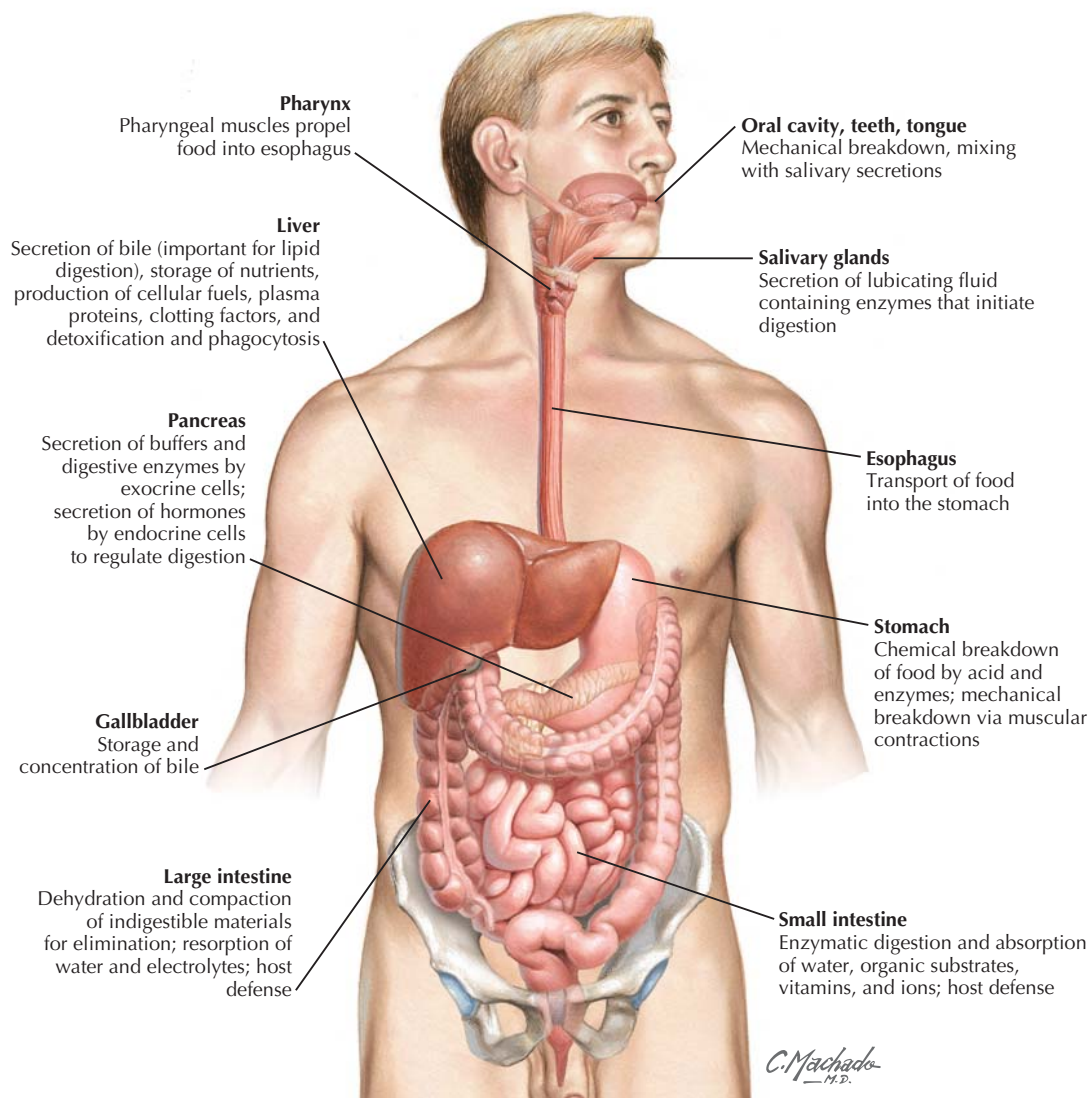


Figure 21.1 Overview of Gastrointestinal Tract The GI tract begins at the mouth and is made up of discrete areas that aid the digestion and absorption of nutrients. Each area of the tract contributes to the efficient processing of the nutrients. This includes the GI-associated organs (liver, pancreas), which provide important secretions including bile, enzymes, and buffers.

Blood Supply

The GI tract has a large blood supply to perform the myriad of functions necessary for digestion, absorption, and propulsion. Oxygen, nutrients, and hormones are brought by the blood to aid in the many secretory, absorptive, and propulsive functions. In addition, efficient absorption of nutrients through the intestinal cells requires a high blood flow, which ensures that there is always a gradient for nutrient entry into the bloodstream.

Blood supply to the GI tract is from the splanchnic bed (superior and inferior mesenteric arteries), with the capillary system



During active digestion, blood flow to the GI tract increases to ~3 times normal to facilitate the metabolic and digestive needs. This rerouting of blood from other tissues (especially skeletal muscle) to the gut can significantly alter the distribution of cardiac output. The proportions of cardiac output perfusing various tissues are described in Section 3 (Chapter 8) for the normal resting state. These proportions are altered postprandially, favoring blood flow to the GI tract.

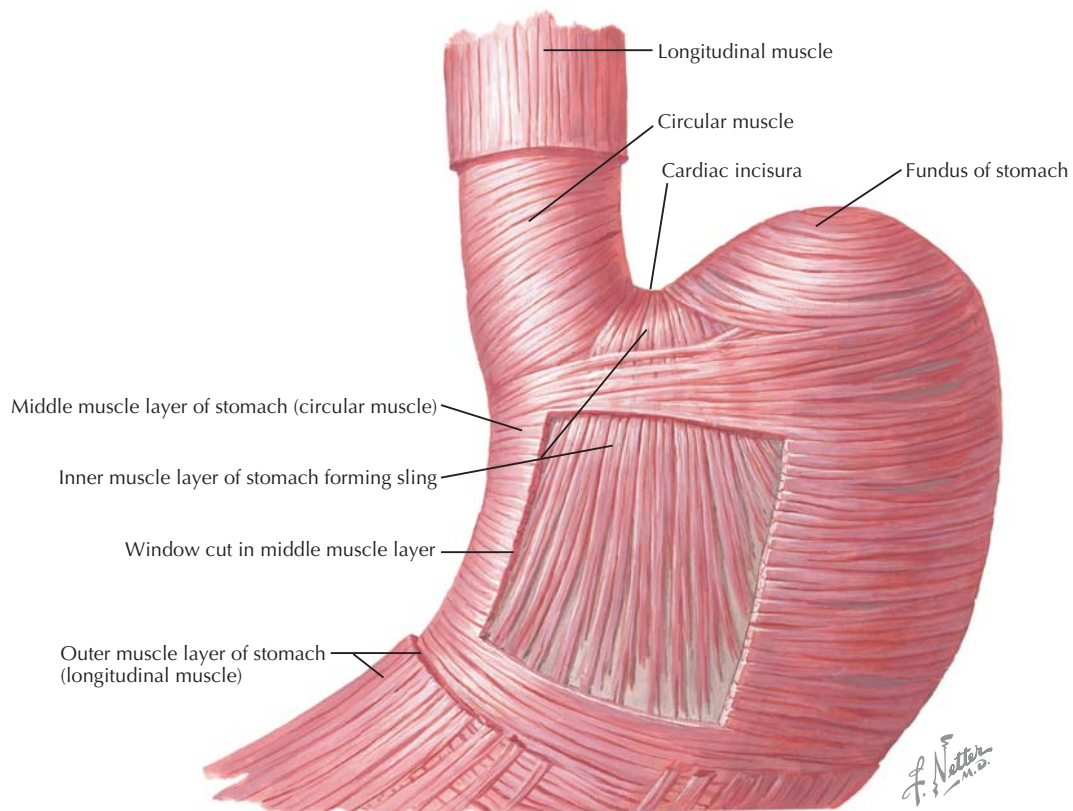


Figure 21.2 Musculature of the Stomach The stomach has longitudinal, circular, and oblique muscle layers. The oblique layer helps mix and grind the chyme.

extending into all of the villi. Blood flows out of the tract through the **portal veins**, directly to the liver (Fig. 21.4). This is the “**first pass**” effect, whereby absorbed substances are routed directly to the liver (without entering into the general circulation), where much processing occurs.

Glandular Secretions

To efficiently move and digest the chyme, a tremendous amount of substances are secreted into the lumen of the tract. Glands produce liters of saliva, mucus, buffers, and enzymes to facilitate the process.

Lymphatics

Lymph lacteals (lymphatic capillaries) are found throughout the intestines and extend into the villi of the small intestine, allowing absorption of lipids. The lymphatic system is also extensive in the liver, where it absorbs fluids and proteins and transports them to the systemic venous blood. Lymphatic transport of proteins contributes substantially to plasma oncotic pressure and provides binding proteins for circulating substances including hormones, calcium, and iron.

General Functions of the GI Tract

- **Digestion:** Both mechanical and chemical digestion occur in the tract. **Mechanical** digestion refers to the physical breakdown of the food by initial chewing and grinding in the stomach. **Chemical** digestion is through specific enzymatic action and gastric acid (HCl).
- **Endocrine:** A great number of hormones are produced by GI-associated cells (from stomach, intestines, pancreas, and liver). Some are hormones, entering the general circulation and affecting other parts of the GI tract as well as other organs. There are also hormones produced that act on adjacent GI tissues in a paracrine manner. In fact, the first substance identified as a “circulating hormone” was the GI hormone **secretin**.
- **Expulsion:** After absorption of nutrients through the tract, the fecal waste matter (fiber, undigested material, bacteria, fat from sloughed cells) is expelled through defecation.
- **Protection:** We constantly ingest bacteria and antigens, and the GI tract keeps this under control through degradation by gastric HCl, as well as immune activity (IgA and Peyer’s patches).

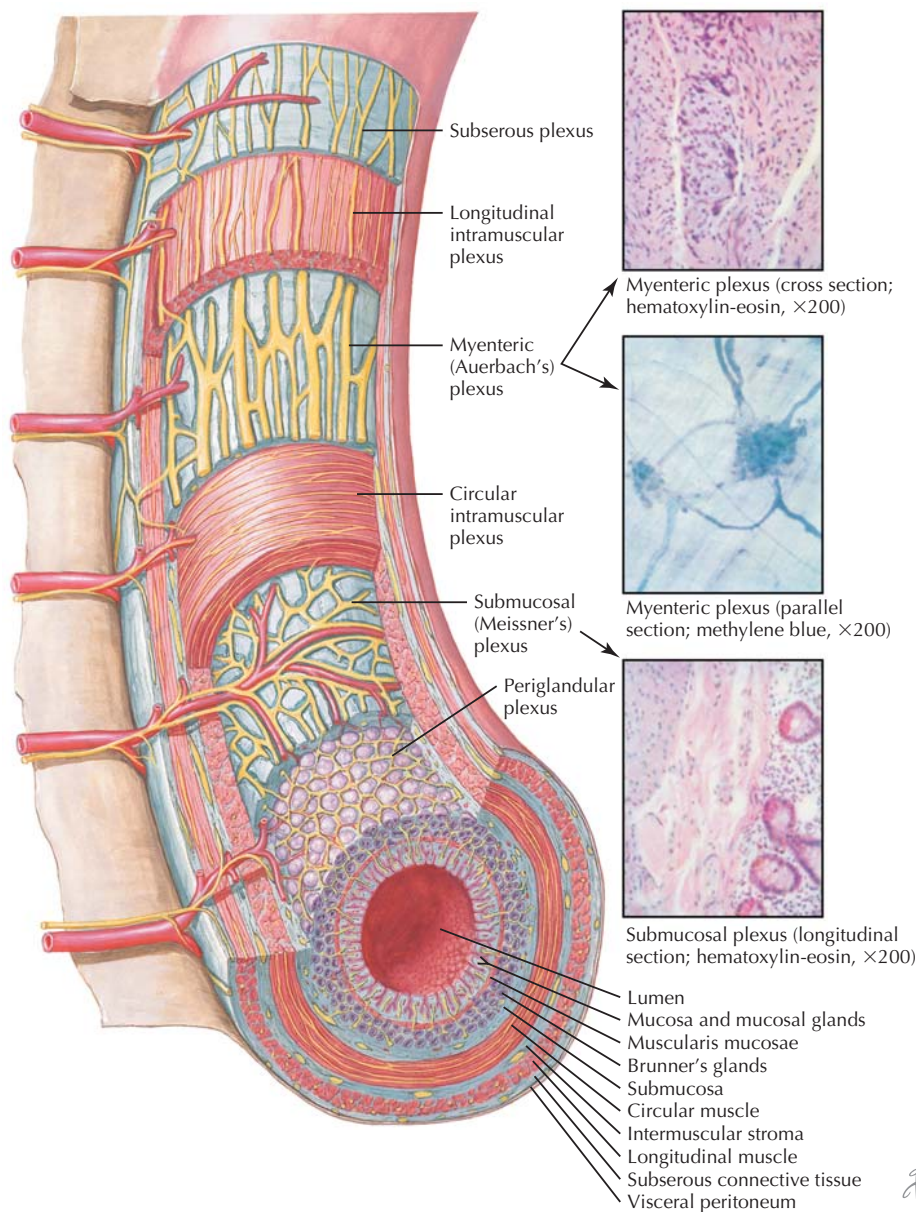


Figure 21.3 The Enteric Nervous System The myenteric plexus (nerve net) is between the longitudinal and circular muscle layers of the GI tract, and stimulation of these nerves affects muscle contraction. The submucosal plexus is between the circular muscle and submucosa, and stimulation can affect GI secretions. The enteric nerves can function without extrinsic nerves, through local mechanisms.

- **Motility:** The chyme needs to be propelled through the tract to allow the digestion, absorption, and expulsion to occur. There are several different forms of propulsion through the tract, and if an area loses the ability to move chyme, serious pathology can occur.
- **Absorption:** The ultimate goal for the elaborate GI system is getting bulk nutrients into the body. The intestines absorb most of the nutrients that are digested, and the majority of absorption occurs in the first half of the

small intestine (through the jejunum). This is accomplished because of the large surface area provided by the villi on the enterocytes of the small intestine.

- **Secretion:** The secretion of mucus, buffers, hormones, and enzymes facilitates lubrication, digestion, motility, and absorption through the tract.
- **Storage:** Both the stomach and large intestine act as storage sites. The stomach accommodates food as it is mixed with the gastric acid and enzymes promoting

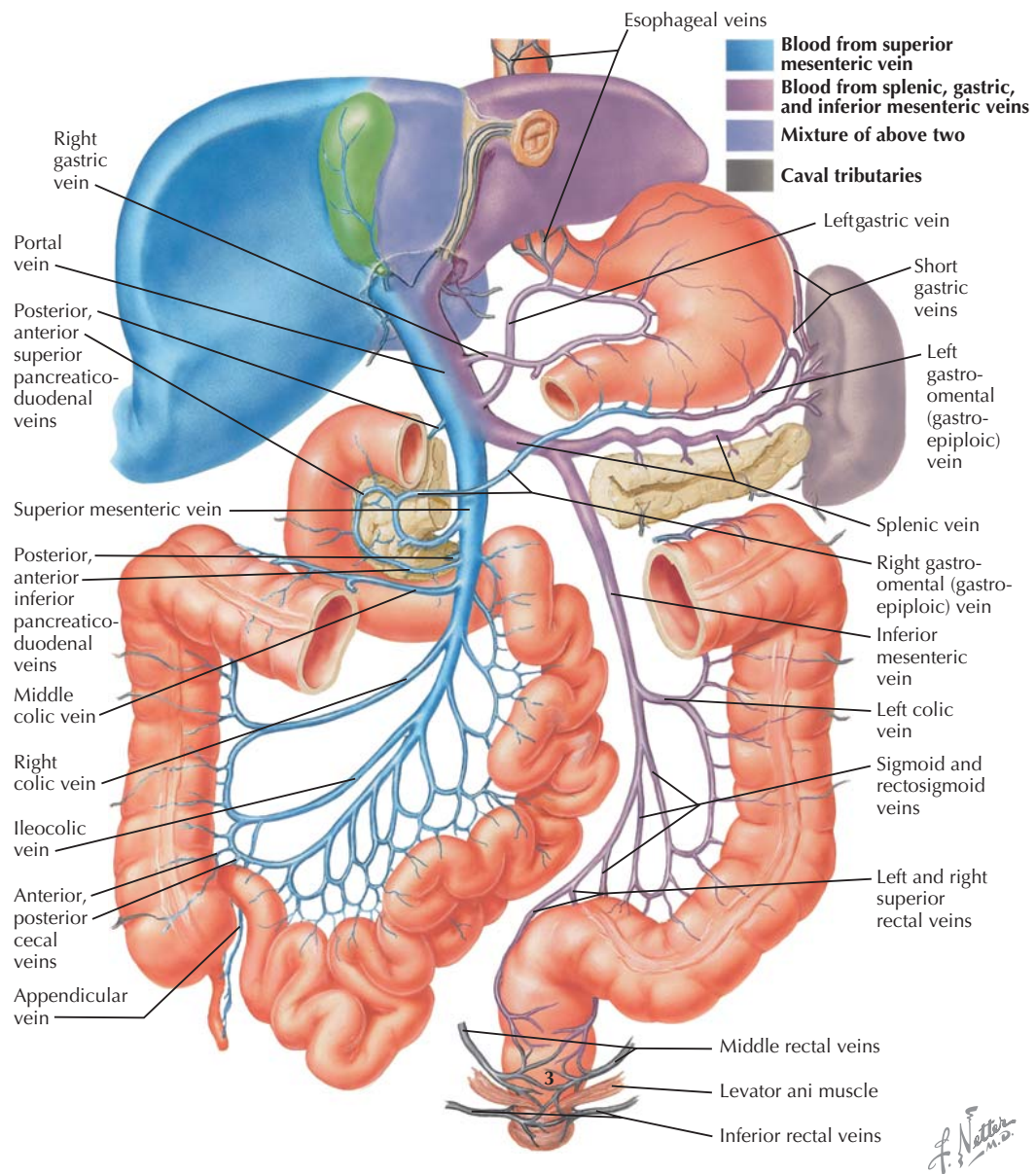


Figure 21.4 The Portal System The portal vessels carry venous blood directly from the intestines to the liver through the portal vein.

breakdown of food into chyme that is small enough to pass through the pyloric sphincter into the duodenum. This storage is facilitated by **receptive relaxation**, mediated by the vagus nerve (see Chapter 22). If the relaxation did not occur, it would dramatically limit the quantity of food that could be ingested during a meal and enter the stomach. On the opposite end, the large intestine dehydrates the chyme to produce feces, and the type of motility there allows retention of the feces until expulsion.

THIRST AND HUNGER: BEHAVIORAL RESPONSES TO MAINTAIN THE MILIEU INTERIEUR

Thirst

To maintain a homeostatic internal environment, we have behavioral responses to ensure adequate ingestion of fluid and nutrients: thirst and hunger. **Thirst** is a response to cell and vascular dehydration causing elevated plasma osmolarity. The

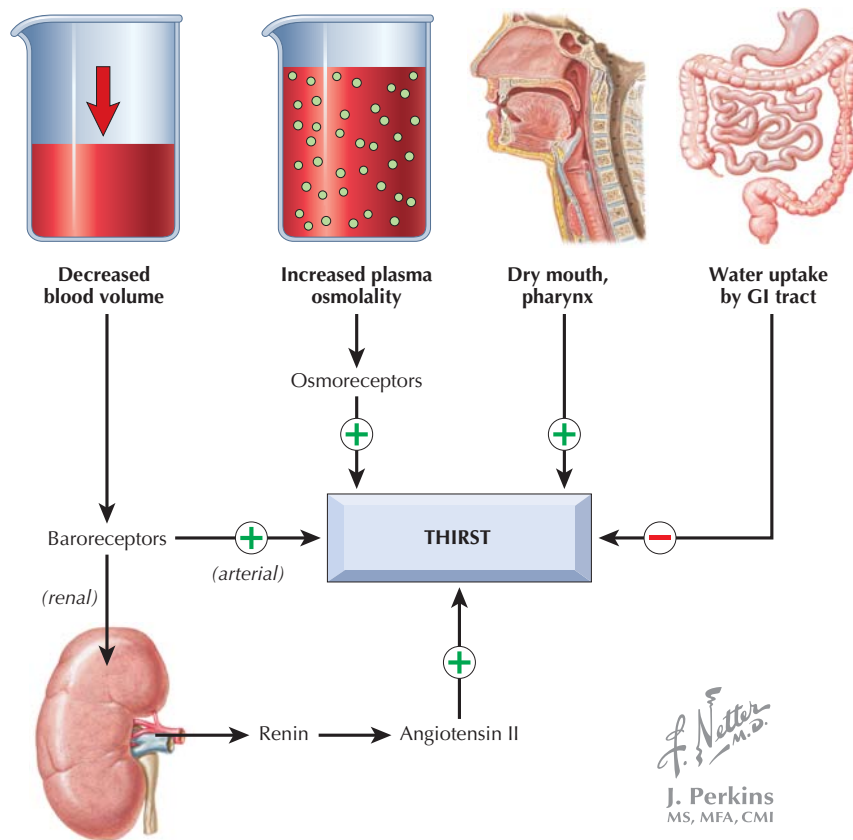


Figure 21.5 Thirst The thirst response is initiated by either a small increase in plasma osmolality (1% to 2%) or a significant decrease in blood volume (~10%). The mechanism is depicted in this illustration. (Modified from Widmaier E, Raff H, Strang K: *Vander, Sherman, and Luciano's Human Physiology*, 9th ed., New York, Mc-Graw Hill, 2003.)

loss of fluid can be a result of urine and fecal fluid loss, as well as **insensible loss** of water from sweating and breathing (Fig. 21.5). The feeling of thirst can be stimulated when hypothalamic osmoreceptors sense elevated plasma osmolality, or arterial baroreceptors sense low blood pressure (often associated with reduced extracellular fluid volume). Both the baroreceptors and osmoreceptors regulate the thirst response (mediated through the hypothalamus) and the release of **antidiuretic hormone** (ADH, also known as **vasopressin**). ADH has effects on renal retention of water (discussed in Section 5) and vascular smooth muscle contraction (discussed in Section 3). Thirst is controlled at both central and local levels. Dry mouth can be relieved temporarily by rinsing the mouth with fluid; however, if the overall cause is dehydration, the thirst response will return rapidly.

Appetite and Hunger

Although the mechanisms are not as clearly understood as those involved in thirst, hunger is a powerful behavioral response to ensure nutrient intake (Fig. 21.6). The main site of control, as with thirst, resides in the hypothalamus, which

has both the hunger and satiety centers. The basic control of the centers is through essential nutrients such as glucose and fatty acids. Stimulation of appetite can occur through low blood glucose, as well as through sensory input (e.g., smelling, seeing food). The GI tract also plays a role, releasing hormones that stimulate (ghrelin, galanin, orexin) and decrease (cholecystokinin, peptide YY) hunger at the level of the hypothalamus. A key factor in satiety is the hormone leptin, which can be released from adipose tissues in response to insulin. Leptin receptors are present on the hypothalamus (as well as other tissues) and leptin stimulates the satiety center. In addition to such specific factors, the hunger



The control of plasma osmolality is so important that a small (2%) change (e.g., from 300 to 306 mosm/L) can induce thirst and ADH secretion. In contrast, a 12% to 15% reduction in extracellular fluid (ECF) volume is needed to stimulate the same ADH response if osmolality is unchanged. Thus, when thirst is stimulated by dehydration, sufficient fluid has been lost to raise plasma osmolality and stimulate ADH.

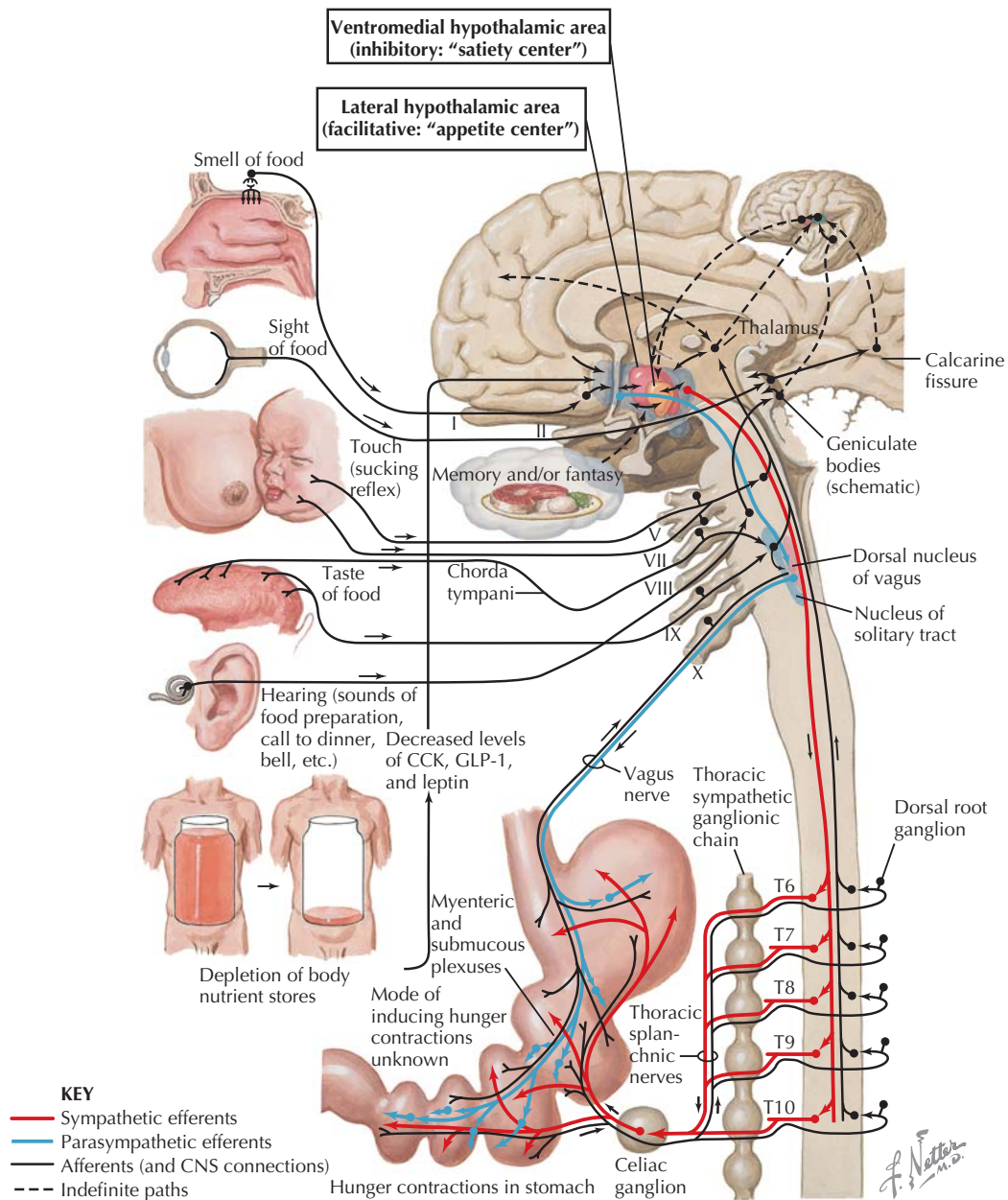


Figure 21.6 Appetite and Hunger The feelings of hunger and satiety are not completely understood; however, general pathways are outlined above. The hypothalamus (see Chapter 26) plays important roles in both hunger (lateral hypothalamic area) and satiety (ventromedial hypothalamic area) and responds to endocrine and nerve input. Multiple hormonal systems facilitate the feeling of satiety: cholecystokinin, peptide YY and GLP-1 (glucagon-like peptide-1) are released in response to ingested food, decreasing hunger. In addition, the hormone leptin is released in response to insulin and acts at the satiety center of the hypothalamus. In contrast, hunger "pains" are elicited through vagal stimulation of the enteric nerves, and the release of other GI hormones, including ghrelin and orexin, stimulate the appetite center of the hypothalamus. CCK, cholecystokinin.



Sweat is hypo-osmotic. Its concentration of salts (mainly sodium and chloride) is less than found in plasma. Thus, sweating depletes the extracellular fluid (ECF) of more fluid than salt, which increases the plasma osmolarity. That stimulates the thirst response and reduces fluid losses by the kidney and colon, helping to maintain blood pressure (see Fig. 21.5).

response is affected by body mass and will provide nutrients for energy expenditure.

FLUID SHIFTS AND pH THROUGH THE TRACT

Although only ~2 liters of fluid is ingested daily, the fluid flux in and out of the GI tract is actually about 9 liters. This is necessary for digestion of the chyme; however, the secretions need to be reabsorbed back into the bloodstream with the added nutrients (Fig. 21.7). With the addition of secretions, the pH of the chyme changes. In the stomach, gastric acid secretion brings the pH to ~2, allowing for indiscriminant digestion of food. To protect the small intestine from the corrosive effects of the acid, buffers (pH ~8) are secreted into the duodenum from the liver, pancreas, and gallbladder, as well as from intestinal crypt cells. This raises the pH of the chyme in the first few centimeters of the duodenum to ~5. The addition of the buffers through the tract brings the pH of the chyme up to ~7.4 by the early jejunum. This is optimal for the pancreatic enzyme action.

While you are conscious of the salivary secretions that occur in anticipation of, and during, feeding, the other fluid fluxes tend to go unnoticed, although at times intestinal sounds called **borborygmi** may be heard—these are the result of fluid and gas fluxes in the lumen.

THE ENTERIC NERVOUS SYSTEM

The GI tract is unique in having an intrinsic nervous system, made up of the **myenteric** and **submucosal plexuses** (see Fig. 21.3). This **enteric nervous system (ENS)** is able to function independently, using input from mechanoreceptors, chemoreceptors, and osmoreceptors located in the luminal epithelium of the tract. The ENS receives input from the central and autonomic nervous system (CNS and ANS) and hormones, which help fine-tune and regulate the ENS. Without autonomic innervation, the ENS would still function, but in a less coordinated manner.

Located between the circular and longitudinal muscle layers, the **myenteric (Auerbach's) plexus** regulates contraction and relaxation of the musculature, causing motility and mixing of the luminal contents. The **submucosal plexus (Meissner's**

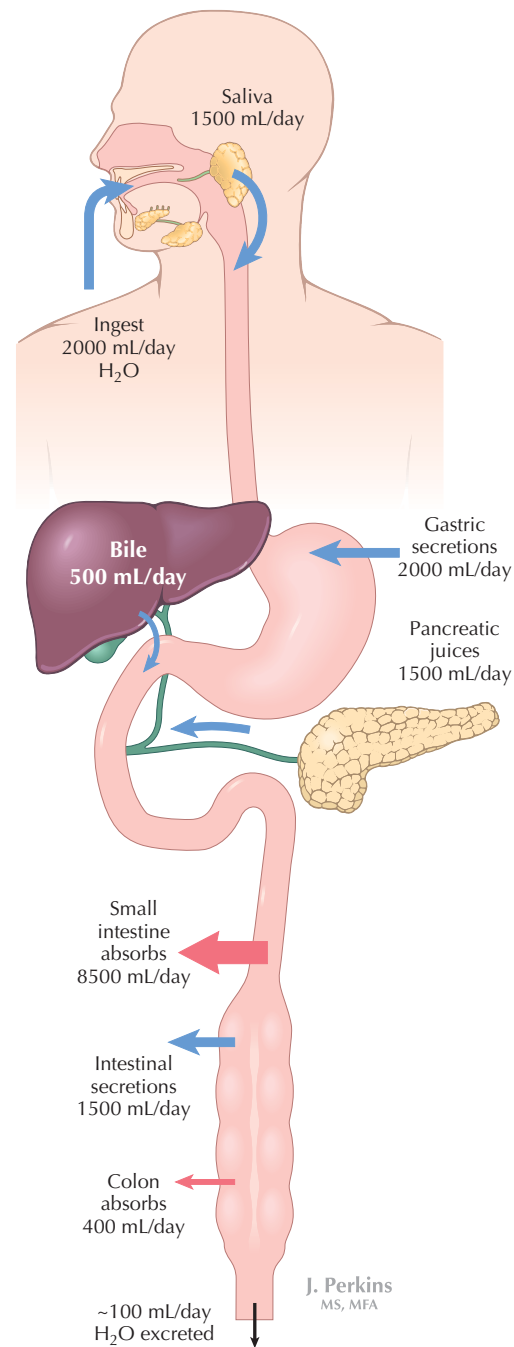


Figure 21.7 Fluid Shifts through the GI Tract While we ingest ~2 liters (L) of fluid in food and drink each day, the GI tract adds ~8 L of secretions to facilitate digestion and absorption of the nutrients. These secretions, including buffers, acid, and enzymes, are absorbed back into the blood (primarily by absorption in the small intestine) so that only ~200 milliliters (mL) of fluid is excreted in feces each day.

plexus) is located between the circular muscle and submucosa and regulates local fluid secretions.

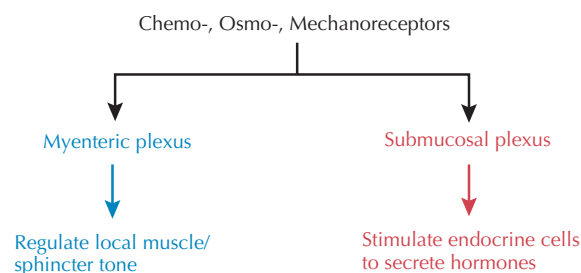
THE GI TRACT AS AN ENDOCRINE ORGAN

Endocrine cells in the gastric and intestinal epithelia synthesize and release a variety of hormones into the bloodstream. The hormones act on other areas of the gastrointestinal system, such as the liver and pancreas, in addition to the stomach and intestines. These hormones regulate GI function, as well as hunger and satiety and insulin secretion. In addition, some hormones act on adjacent cells in a paracrine manner. An example of this is the action of somatostatin (produced in gastric pit cells) on adjacent parietal cells. GI hormones will be covered in Chapter 23.

INTEGRATED REGULATION OF GI FUNCTION

Proper GI function results from the integrated responses from central and local factors including:

- **Central nervous system (CNS):** Sensory input to the CNS provides initial stimulus for salivary and gastric acid secretion and is integral to many reflexes. Simply smelling or seeing food will initiate a central response. The CNS acts through the sympathetic and parasympathetic nervous systems.
- **Enteric nervous system (ENS):** The myenteric and submucosal nerve plexuses are unique to the GI tract. These nerve nets receive input from the PNS and SNS, as well as hormones, peptides, and luminal receptors that sense the chemical composition of the chyme.
- **Parasympathetic nervous system (PNS):** In general, the PNS promotes secretion and motility in the GI tract, and many of the actions are through the vagus nerve.
- **Sympathetic nervous system (SNS):** The SNS slows secretion and motility in the GI tract. Think of what happens in “fight or flight” reactions, in which you need more blood flow to skeletal muscle, and require less perfusion of the intestines.
- **Luminal receptors:** The GI tract has many different receptors that react to the chyme in the lumen of the tract and act locally on the ENS to regulate motility and secretions through the general scheme:



- **Mechanoreceptors** sense stretch of the smooth muscle, and the generated signal is transduced through the myenteric plexus, stimulating contractions.
- **Chemoreceptors** sense the chemical composition of the chyme and regulate motility and secretion of buffers to control luminal pH during the influx of acidic chyme into the duodenum.
- **Osmoreceptors** sense the osmolarity of the chyme in the small intestine. This is important, because there is only a one-cell barrier between the chyme in the lumen of the small intestine and the capillaries, and hypertonic chyme can exert an osmotic force, pulling fluid out of the cells (and ultimately the plasma). Thus, the osmoreceptors control the amount of chyme entering the small intestine, as well as the amount of secretions necessary to buffer the chyme.
- **Hormones:** As stated above, the GI tract produces a variety of endocrine, paracrine, and autocrine factors, which increase the efficiency of the tract (both for motility and secretions).

Lastly, while all of these areas are important in the coordinated function of the GI tract, it bears noting that the **parasympathetic nerves**, primarily via the **vagus**, are responsible for multiple effects early in the digestive process. Some of the actions they stimulate or participate in include the following:

- Initiating salivation upon seeing, smelling, tasting food during the cephalic phase of salivation (through facial and glossopharyngeal nerves).
- Initiating acid production when food is in the mouth during the cephalic and gastric phases of gastric acid secretion (vagus).
- Stimulating pancreatic enzyme secretion during the cephalic and gastric phases of pancreatic secretion (vagus).
- Stimulating pancreatic enzyme and buffer secretion during the intestinal phase of pancreatic secretion (vagus).
- Stimulating primary (swallowing center of medulla and vagus) and secondary esophageal peristalsis (vagus, with enteric nerves).
- Initially relaxing the sphincter of Oddi during cephalic and gastric phases (vagus).
- Causing receptive relaxation of stomach and duodenum (vagus), to accommodate entry of food/chyme.
- Stimulating the synthesis of bile by the liver (vagus).
- Stimulating intestinal motility (ileal motility and colonic mass movements) (vagus to upper colon, pelvic nerves on lower colon; with gastrin and cholecystokinin [CCK]).

Thus, if vagal innervation of these areas is impaired, the digestive and propulsive processes will be dysregulated.

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CHAPTER 22

Motility through the Gastrointestinal Tract

ELECTRICAL POTENTIALS

Motility in the gastrointestinal (GI) tract involves both mixing and propulsive movements, and is determined by the mechanical activity of the intestinal smooth muscle. The movements serve to mix, transport, and eliminate matter. To accomplish the muscle contraction, electrical potentials must be generated.

The electrical activity in the GI tract is unique. Unlike other tissues, there are undulations in the resting membrane potential known as slow waves, which are generated by the interstitial cells of Cajal (ICC). The ICC are located throughout the GI tract between the longitudinal and circular muscle layers, and the undulations in their resting potential are probably caused by the small changes in membrane charge resulting from Na^+/K^+ ATPase activity. The ICC act as pacemakers, determining the number of waves that occur per minute along different segments of the GI tract. Because of gap junctions between cells, the slow waves (also called the **basic electrical rhythm [BER]**) can be propagated over relatively long segments of the tract.

Under resting conditions, the slow waves undulate between -70 and -80 millivolts (mV) and do not cause contractions. However, if the slow waves are depolarized (made less negative) by nerve activity or circulating hormones, the amplitude of the waves may increase, and when peaks of the slow wave cross the threshold of -40 mV, the cells will generate one or more action (or spike) potentials.

Neurotransmitters such as acetylcholine (ACh) and substance P (tachykinin) released from parasympathetic nerves terminating on the myenteric plexus depolarize the slow waves, generating action potentials and causing contractions (Figs. 22.1 and 22.2). With greater neurotransmitter release, the amplitude of the waves is higher, producing greater depolarization. This results in more action potentials, increasing the strength of the contraction. Some GI hormones (cholecystokinin [CCK], gastrin) can also depolarize the slow waves, causing contractions. In addition, local mechanoreceptors sensing stretch or chemoreceptors sensing composition of chyme signal the myenteric plexus to fire excitatory motor neurons, depolarizing the slow waves and causing contraction



The action potentials (AP) generated by depolarization of the slow waves are caused by the entry of calcium into the smooth muscle via voltage-gated channels. The calcium binds to calmodulin, initiating events that result in smooth muscle contraction.

Table 22.1 Slow Waves

Area	Frequency of Slow Waves	General Actions when Depolarized above Threshold
Stomach	3/minute	Mixing
Duodenum	12/minute	Propulsion
Ileum	10/minute	Propulsion
Proximal Colon	3/hour	Haustra formation and storage
	11/minute	Mass movements and propulsion
Distal Colon	10/hour	Haustra formation and storage
	17/minute	Mass movements and propulsion

(Fig. 22.3). The characteristic frequency of the slow waves for any section of the tract determines the maximum rate at which contractions could occur, and thus the BER sets the maximal rate of propulsion possible (Table 22.1).

There are also inhibitory motor neurons that are stimulated by the sympathetic nerves and release **vasoactive intestinal peptide (VIP)** and nitric oxide (NO), which hyperpolarize the slow waves (make them more negative), relaxing the smooth muscle (see Fig. 22.3). The interplay between excitatory and inhibitory motor neurons results in a variety of propulsive and mixing movements through the tract (see “Site-Specific Propulsion”).

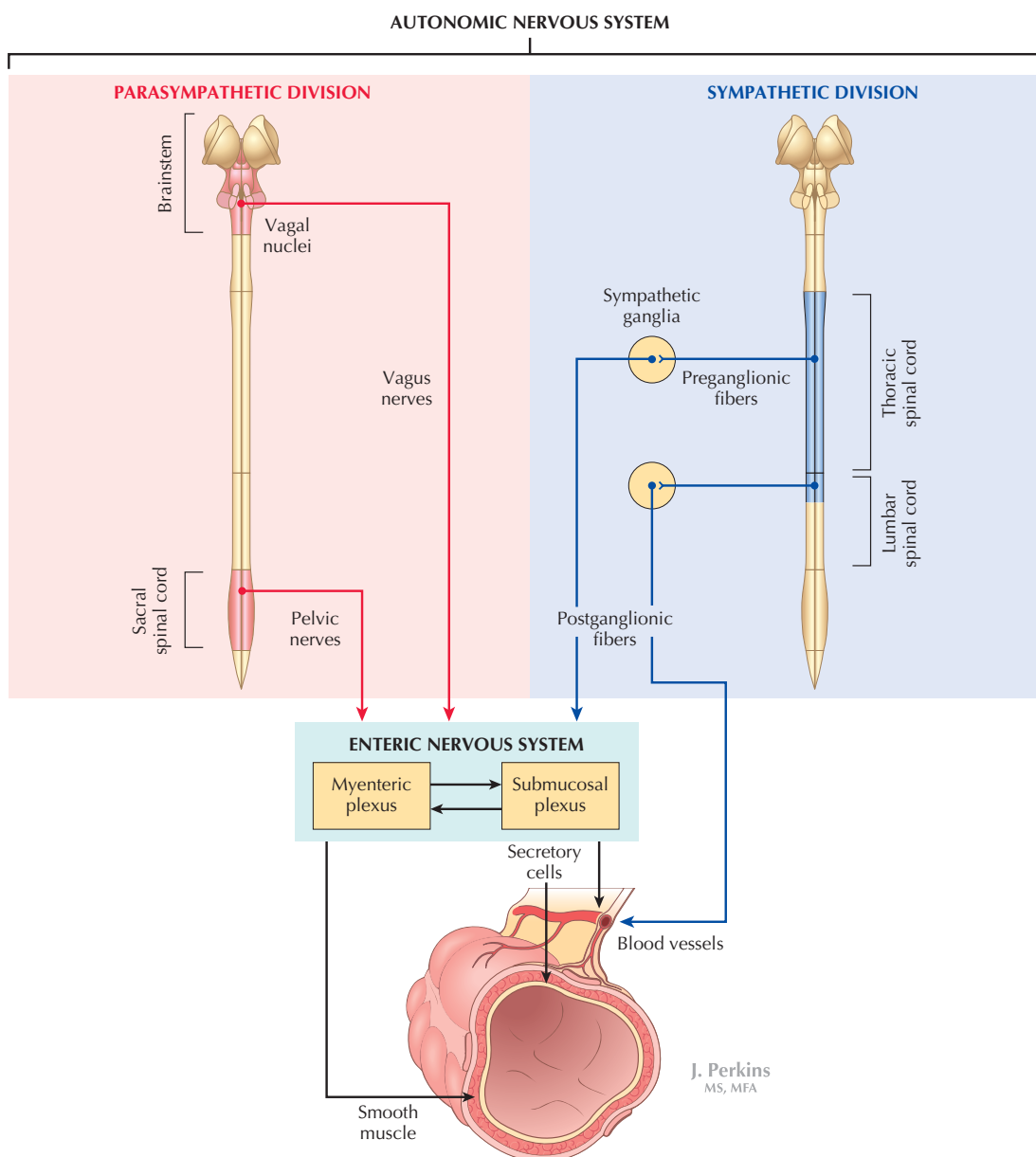


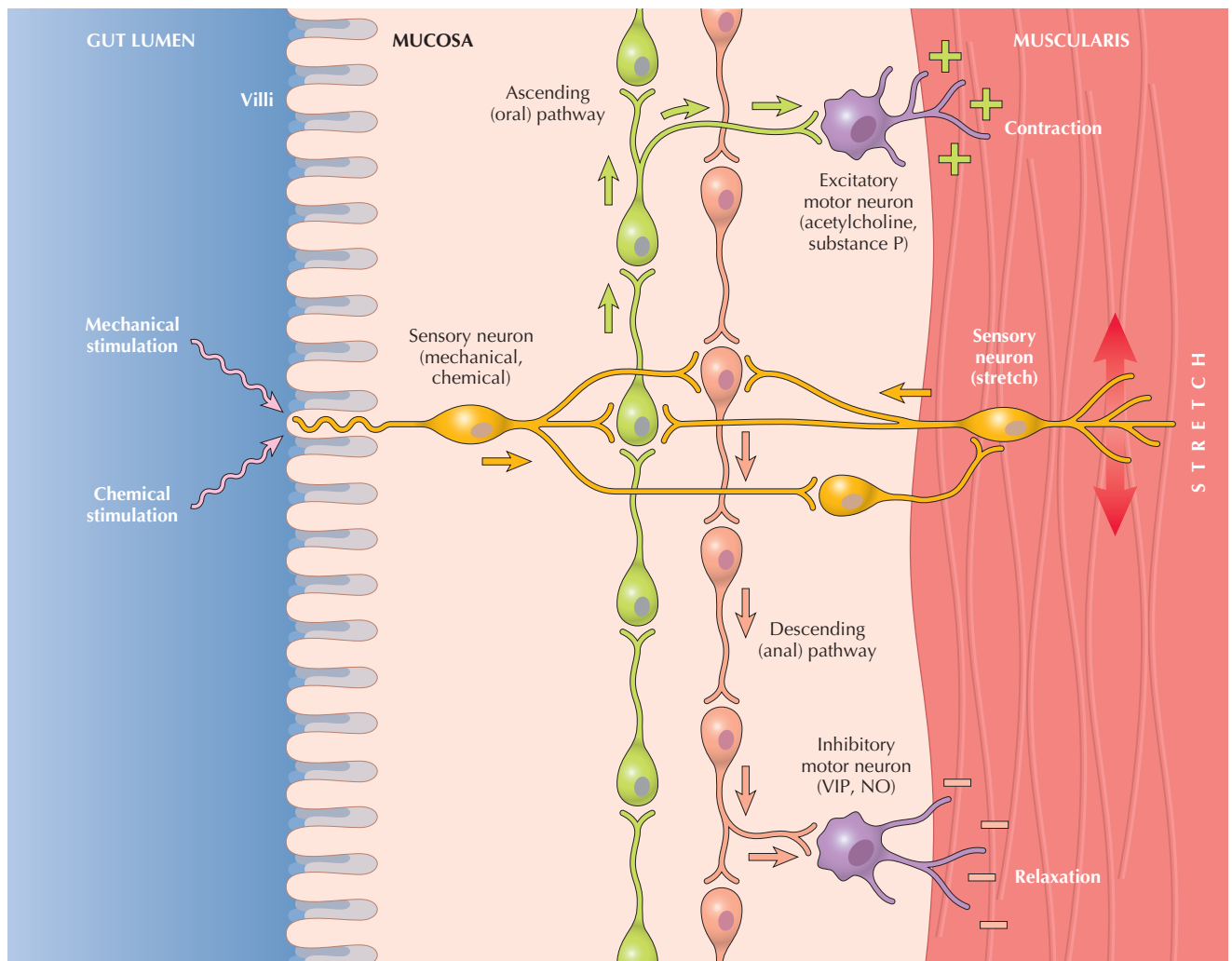
Figure 22.1 Regulation of Enteric Nervous System by Autonomic Nervous System This diagram illustrates basic connections between the enteric nervous system (myenteric and submucosal plexuses) and the autonomic nervous system. In general, stimulation from the parasympathetic nerves enhances motility and secretions through the ENS, whereas sympathetic stimulation reduces motility and secretions.

INTERDIGESTIVE HOUSEKEEPING: THE MIGRATING MYOELECTRIC COMPLEX

Fasting, or the interdigestive state, is characterized by long periods of quiescence, with short periods of waves of contractions. The contractions, called the **migrating myoelectric complex (MMC)**, originate in the mid-stomach and continue to the terminal ileum during each cycle and serve to sweep the undigested material and bacteria out of the stomach and small

intestine, into the colon. This protects the delicate upper GI tract (small intestine) from damage and sequesters most of the bacteria in the colon, a drier environment less hospitable to bacteria.

There are four phases of the MMC that occur in 75- to 120-minute cycles during fasting, based on contractile activity. Phases I, II, and IV have little activity. Phase III is most



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Figure 22.2 Local Control of Motility In response to the presence of chyme in the lumen of the small intestine, mechanoreceptors and chemoreceptors transduce signals to ascending and descending neurons. The ascending (*green*) pathway leads to excitatory motor neurons behind the bolus of chyme, stimulating depolarization of the slow waves, generation of an action potential, and contraction. At the same time, the descending (*orange*) pathway ends in inhibitory motor neurons, which hyperpolarize the slow waves in the muscle, causing relaxation in front of the bolus. The overall result is a peristaltic contraction, moving the bolus toward the anus.

important of the four, when the hormone **motilin** is released from Mo cells of the small intestine into the circulation. Motilin stimulates strong sequential contractions, sweeping bacteria and undigested matter down the tract from mid-stomach into the colon. Phase III lasts only 6 to 10 minutes in each cycle. Motilin acts through the enteric and autonomic nerves to stimulate contractions, which proceed for a few feet aborally, and then start up again slightly further down the intestine. This pattern is repeated until the complex reaches the terminal ileum, and then a new complex is initiated in the stomach.

These cycles begin about 3 hours after the last meal and repeat until ingestion of food. With ingestion, the normal propulsive patterns resume.

SITE-SPECIFIC PROPULSION

Mouth and Esophagus

Food is first moved by chewing and then **swallowing**. There are three stages of swallowing.

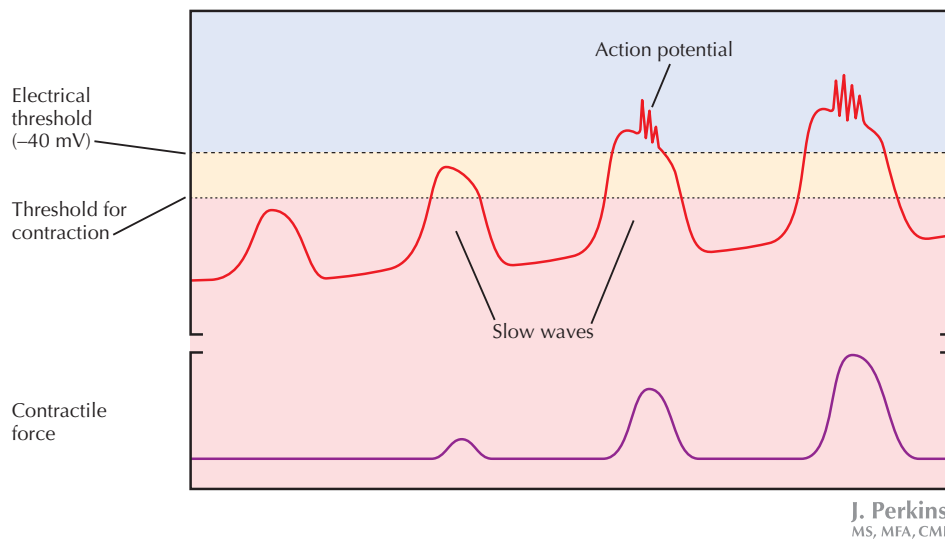


Figure 22.3 Slow Waves Slow waves are present from mid-stomach through the rectum and are the resting membrane potential of the GI smooth muscle. Depolarization of the slow waves above -40 mV stimulates action potentials, which causes contraction of the smooth muscle.

In the voluntary **oral** stage, the bolus of food is moved to the back of the mouth into the pharyngeal region, stimulating touch receptors and initiating the swallowing reflex (Table 22.2). In the **pharyngeal** stage, the bolus of food moves to the back of the pharynx, the larynx moves toward the epiglottis to prevent food entering the trachea, and the upper esophageal sphincter relaxes. The pharyngeal muscles contract, propelling the bolus through the upper esophageal sphincter into the esophagus. During the pharyngeal phase, the swallowing reflex prevents respiration. The last is the **esophageal** stage. When the bolus enters the esophagus, voluntary control of the movement is lost. A wave of **primary esophageal peristalsis** is generated by the swallowing center in the medulla (efferent vagal nerves terminate on the myenteric plexus $\rightarrow$ excitatory signal $\rightarrow$ depolarized slow waves $\rightarrow$ contraction). The alternating contraction and relaxation helps carry the bolus of food through the esophagus to the stomach.

During swallowing, if the bolus is dry and does not move quickly through the esophagus, **secondary esophageal peristalsis** is initiated by local mechanoreceptors sensing stretch. The **lower esophageal sphincter** has a high resting muscle tone, to prevent reflux of gastric acid. The basal tone is maintained by enteric nerves, hormones, and vagal cholinergic fibers: The tone can be increased by sympathetic stimulation. As the bolus nears the lower esophageal sphincter, noncholinergic vagal fibers acting on inhibitory interneurons release VIP and NO, causing relaxation of the sphincter (Fig. 22.4). The bolus then passes into the stomach. If the lower esophageal sphincter is unable to relax, pathology can occur (see “Smooth Muscle Disorders: Achalasia and Hirschsprung’s Disease” Clinical Correlate). In addition, the esophagus, like the rest of the tract, is highly vascularized, and damage to the

esophageal capillaries (acid reflux, portal hypertension) can result in esophageal bleeding, which can be life-threatening (see “Portal Hypertension and Esophageal Varices in Obstructive Liver Disease” Clinical Correlate in Chapter 24).

Stomach

As the bolus enters the stomach, **receptive relaxation** increases the stomach size, allowing accommodation of the meal. This is vagally-mediated through the release of VIP. As the stomach fills, the stretch and chemical contents stimulate contractions, which help mix the food with the gastric secretions, forming chyme. As the chyme mixes, the contents separate, with carbohydrates and readily digested substances in the lower part (antrum), large chunks in the body of the stomach, and most fats floating at the top. The **transit time** through the stomach depends on the amount and type of food ingested. Small, easily digested meals (high in carbohydrates such as pasta and sugar) move through the stomach quickly (30 to 60 minutes). Meals with more solid foods (meats) and high fat content (fried foods) take much longer (3 to 4 hours) to clear the stomach.

Figure 22.5 illustrates the waves of contractions that begin mid-stomach and move the chyme toward the antrum and pylorus (which acts as a sphincter). Regulation of the tone of the pyloric sphincter occurs through both neural and hormonal pathways (Table 22.3).

For the most part, pyloric tone is high, so as the stomach is filling and contractions are starting, most of the acidic chyme moving into the antrum undergoes retropulsion away from the sphincter (see Fig. 22.5B, 1 and 2). As the waves of

CLINICAL CORRELATE

Bariatric Surgery

Receptive relaxation is crucial for the storage function of the stomach, which allows the contents to be well mixed with gastric acid and enzymes and some digestion to occur. If vagal innervation to the fundus is lost, the accommodation will not occur, and stomach pressures will quickly rise, imparting a feeling of “fullness,” and the inability to physically get more into the stomach. Reducing the storage ability of the stomach is the rationale behind gastric bypass and stomach stapling in severely obese persons.

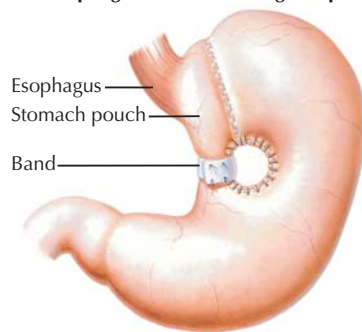
- **Adjustable gastric banding** restricts the size of the pylorus, so that only small amounts of food can comfortably enter the stomach. This restricts food intake because of discomfort, but does allow food to enter the intact stomach.
- **Vertical banded gastroplasty** was a common procedure involving both stomach stapling and banding, to create a small pouch

while leaving the rest of the tract intact. It is considered dangerous and is performed less frequently today.

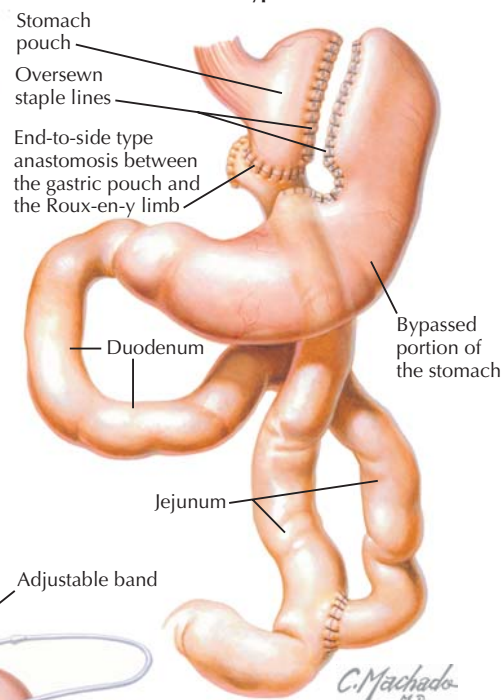
- **Gastric bypass**, a common surgery, involves keeping only a small part of the fundus and attaching it to the early jejunum. This leaves little space for food (about a tablespoon-full). In this procedure, the main body of the stomach is “stapled” off and the duodenum is reattached to the jejunum, allowing the digestive juices (primarily from the pancreas and liver) to mix with the food. The overall effect is a reduction in food intake, and weight loss.

Bariatric surgery is not without risk and can be associated with vomiting, diarrhea, reflux, leaking from the surgical sites, and infection. It is usually only recommended in morbidly obese patients.

Gastric stapling (vertical banded gastroplasty)



Gastric bypass (Roux-en-Y)



Laparoscopic adjustable gastric banding

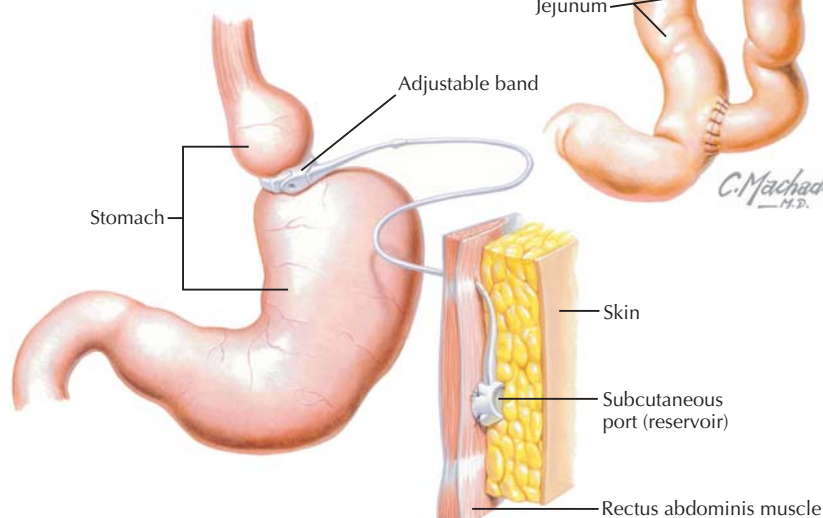


Table 22.2 Main Reflexes through the GI Tract

Reflex	Action	Mediated By
Swallowing	Contraction of pharynx and upper esophagus, inhibition of respiration	Touch receptors in pharynx to swallowing center of the medulla; back to pharynx and upper esophagus through cranial nerves, and to the rest of the esophagus through vagal nerves
Reflexive relaxation of stomach and duodenum	Relaxes the fundus and body of stomach when food, water, or gas is present. Also occurs in duodenum when bolus enters.	Vagal fiber release of VIP
Vomiting	Expulsion of upper intestinal and gastric contents by reverse peristalsis	Irritation of pharynx, esophagus, stomach, or intestines, via vagal and sympathetic afferents to vomiting center; or stimulation of chemoreceptor trigger zone (in medulla near area postrema) to vomiting center
Peristalsis	Contraction behind bolus, relaxation ahead of bolus	Mechanoreceptor and hormone action on enteric nerves
Gastrocolic	Mass movements in colon after meal	PNS and hormones (CCK, gastrin)
Gastroileal	Increased segmentation in ileum in response to gastric emptying	PNS and hormones (CCK, gastrin)
Ileogastric	Chyme in the ileum decreases gastric emptying	Enteric and autonomic nerves
Enteroenteric (aka Intestinointestinal)	If one area of SI is overdistended (e.g., by bacterial infection) the rest of the SI will relax, causing cessation of motility	Enteric and autonomic nerves
Colonocolonic	Distension of one area of colon will relax other areas	Enteric and sympathetic nerves
Rectosphincteric (aka Defecation Reflex)	Feces entering the rectum will cause peristalsis and relax the internal anal sphincter	Both local (enteric nerves) and PNS

These reflexes do not act in an all-or-nothing manner; many act at the same time to create efficient movement of the chyme.

VIP, vasoactive intestinal peptide; PNS, parasympathetic nervous system; CCK, cholecystokinin; SI, small intestine.

Table 22.3 Effects of Nerves and Hormones on Gastric Emptying

Effector	Action on Pyloric Sphincter
Sympathetic nerves	Constriction
Parasympathetic vagal nerves Excitatory via ACh motor neuron Inhibitory via VIP motor neuron	Constriction Relaxation
Hormones—gastrin, GIP, CCK, secretin	Constriction

ACh, acetylcholine; VIP, vasoactive intestinal peptide; GIP, glucose insulinotropic peptide; CCK, cholecystokinin.

contraction progress into the antrum, the **antral cycle** occurs, with relaxation of the sphincter and ejection of some chyme into the duodenal bulb (see Fig. 22.5B, 3). This cycle (ejection into duodenum) is under tight control, and as chyme enters the antrum and duodenum, the release of hormones decreases gastric emptying by constricting the sphincter (see Table 22.3). Thus, relaxation via vagal inhibitory signals allows the antral cycle to occur, while the other factors prevent too much acidic chyme from entering the duodenum at once.

Small Intestine

When chyme enters the duodenal bulb, receptive relaxation also occurs via the vagus nerve. In the small intestine, there are

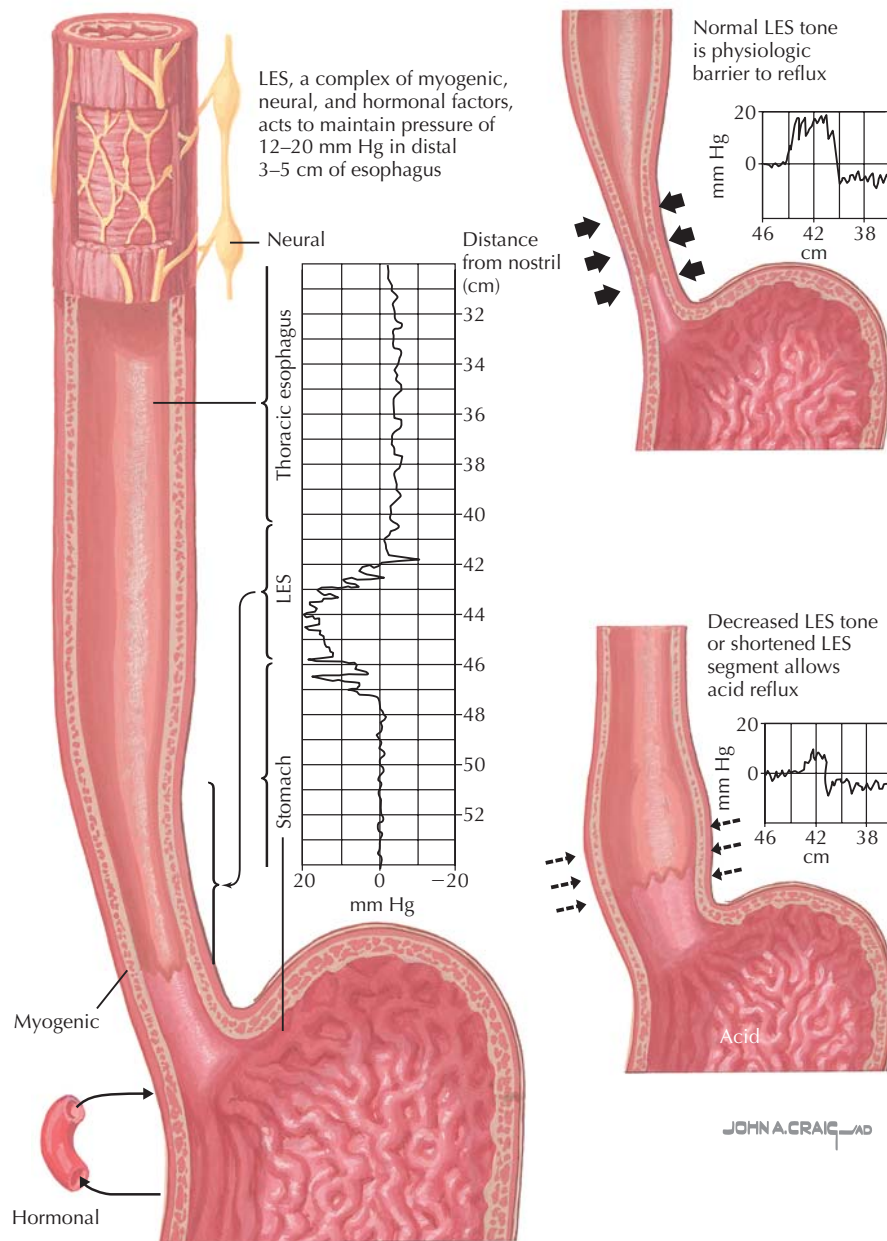


Figure 22.4 Lower Esophageal Sphincter The resting tone of the lower esophageal sphincter (LES) is usually very high, preventing reflux of stomach contents. When food is swallowed, esophageal peristalsis is initiated by the vagus nerve and is propagated by the enteric nerves. As the bolus of food reaches the LES, local nitric oxide (NO) and vasoactive intestinal peptide (VIP) are released, and the sphincter tone relaxes, allowing the bolus to enter the stomach. If LES tone is decreased when at rest, acid reflux can occur.

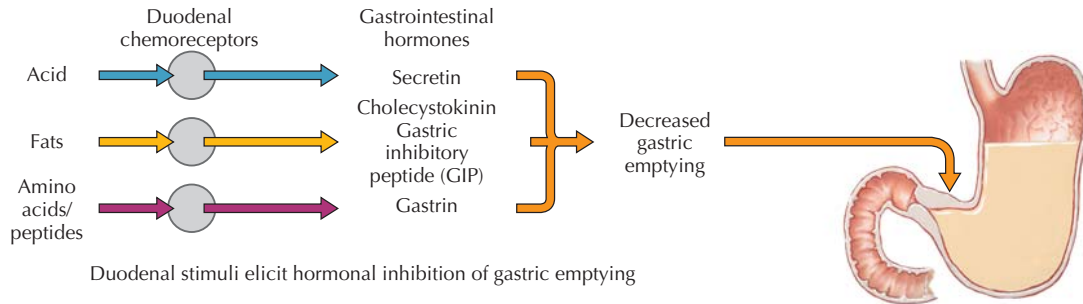
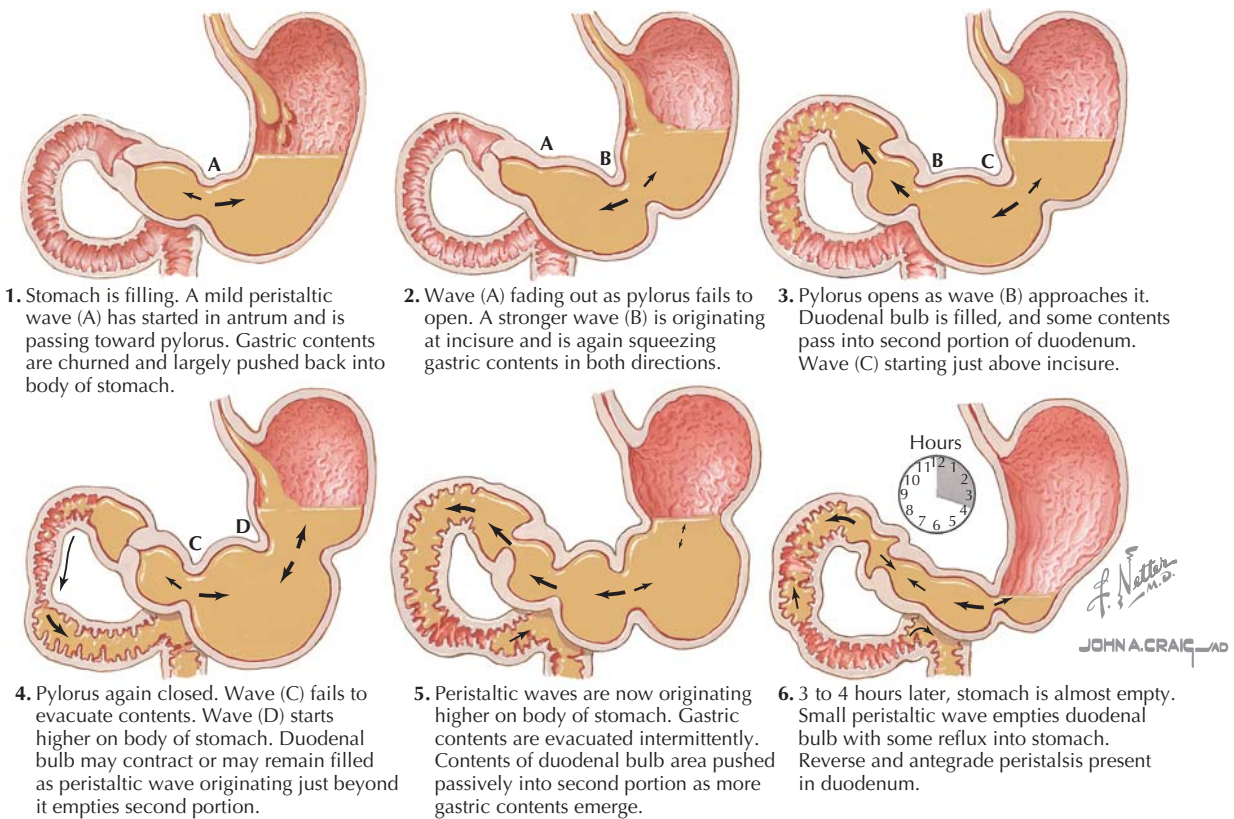
A. Factors affecting gastric emptying**B. Sequence of gastric motility**

Figure 22.5 Gastric Motility The presence of food in the stomach stimulates gastric motility through release of vasoactive intestinal peptide (VIP) stimulated by vagal nerves. The stomach contractions force the gastric contents toward the pylorus, where most of the contents are pushed back into the body of the stomach (1). As spurts of chyme enter the duodenum (3), duodenal hormones are released into the blood, circulate back to the stomach, and decrease gastric emptying (A). Thus, there is a coordinated effect of local and autonomic nerves, and hormones, controlling gastric motility: The entire sequence is depicted in B, images 1–6.

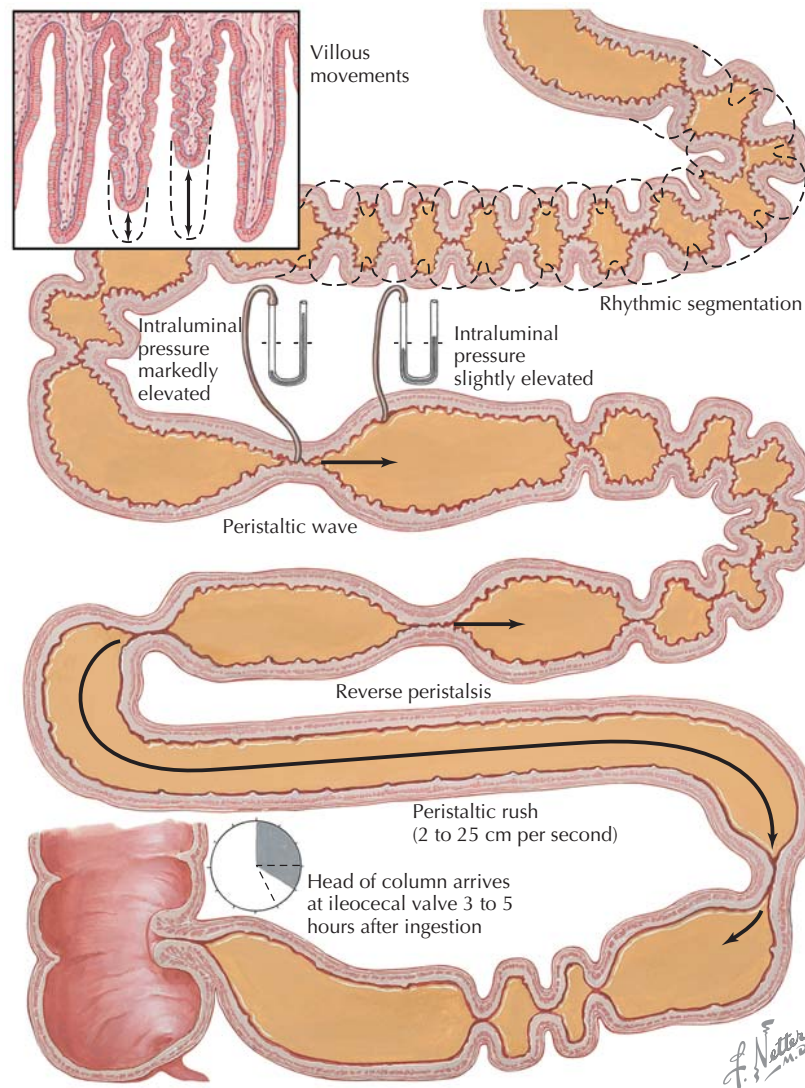


Figure 22.6 Peristalsis and Segmentation Motility in the small intestine is primarily under local control of the myenteric plexus and consists of both peristalsis and segmentation. Segmentation forms pockets of chyme and serves to mix and propel the chyme, although normally peristalsis creates aboral movement (away from mouth). Peristaltic rushes can occur when the intestines are irritated, moving chyme rapidly through the intestines.

two types of propulsion, peristalsis and segmentation. **Peristalsis** follows the “law of the intestines,” according to which contraction takes place behind the bolus of chyme and relaxation ahead of the chyme (Fig. 22.6). This process is accomplished by the simultaneous stimulation of excitatory and inhibitory motor neurons (see Fig. 22.2). In this manner, the chyme is propelled aborally (away from the mouth). In the small intestines, there can also be **peristaltic rushes**, which quickly propel the chyme over long segments. These occur when there is irritation or bacteria in the segments, and the movement facilitates removal of the irritant. Because this quickly sweeps the chyme through the intestines, much less absorption occurs, and this can result in **diarrhea** (see Fig. 22.6). **Reverse peristalsis** can

also occur (seen in vomiting), rapidly moving intestinal and stomach contents toward the mouth.

Segmentation is the formation of moving pockets of chyme, by close constriction of circular muscle (see Fig. 22.6). Unlike peristalsis, which contracts behind the bolus, segmentation contracts in the middle of the bolus, spreading the chyme proximally and distally. These constrictions occur rhythmically and are the most frequent type of contractions in the small intestine. Because the simultaneous constriction moves in waves, the “pockets” slowly move aborally, mixing contents backward and forward. Although much mixing occurs, there is also net aboral movement. Segmentation and peristalsis

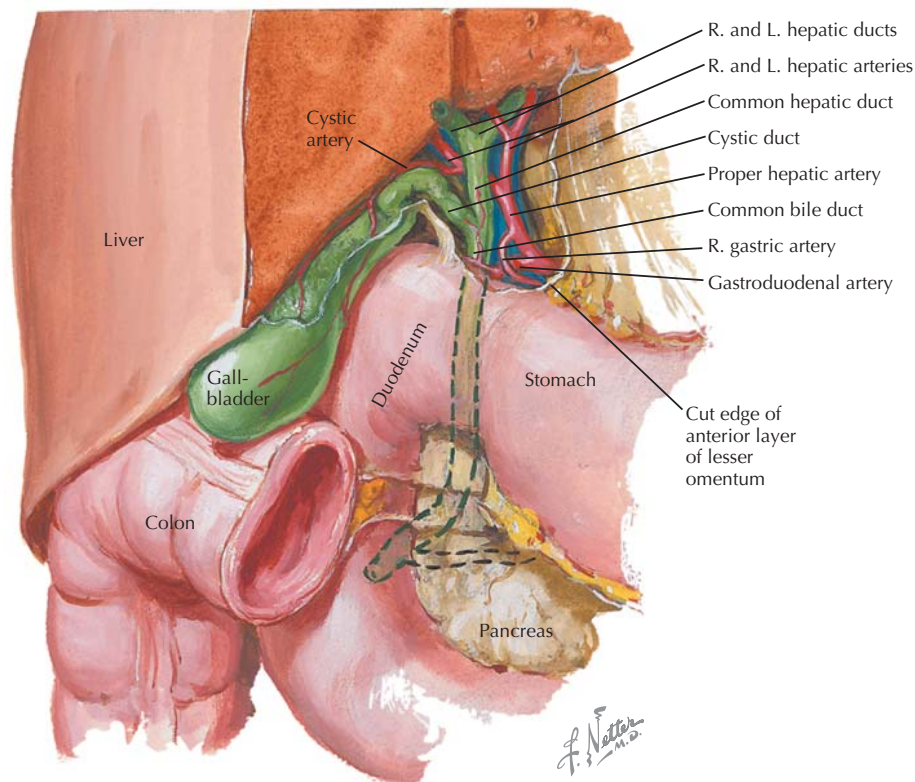


Figure 22.7 Structure of the Gallbladder The gallbladder is the storage site for bile and is located adjacent to the liver. It can store between 25 and 50 milliliters (mL) of bile, which is released by vagal stimulation and cholecystokinin. When food and chyme are in the stomach, the vagus stimulates initial contraction. Subsequently, when chyme enters the duodenum, cholecystokinin is released, causing strong contractions and emptying of the gallbladder. Bile is concentrated (salts and water removed) in the gallbladder, allowing more storage.

occur in adjacent segments of the small intestine throughout the digestion and absorption of the chyme. This pattern is orchestrated by the myenteric nerves (with fine-tuning by autonomic nerves and hormones).

Gallbladder

The gallbladder is a specialized structure for concentration and storage of bile during the interdigestive state (Fig. 22.7). When chyme first enters the duodenum, cholecystokinin (CCK) is secreted into the blood (in response to the presence of fats and glucose in the chyme). CCK, in combination with vagal stimulation, causes contraction of the gallbladder, forcing bile out into the cystic duct and the common bile duct. CCK (and vagal stimulation) also relaxes the **sphincter of Oddi**, allowing the contents of the common bile duct to enter the duodenum.

Large Intestine (Colon)

The colon also has specialized forms of propulsion: segmental propulsion and mass movements. The muscle structure of the

large intestine is different from the small intestine, in that there are three bands of longitudinal muscle, called the **taenia coli**, which run the length of the organ. When they contract, they form sacs called **hastra**; this is **segmental propulsion** (Fig. 22.8). The hastra stay in formation for long periods, which helps “store” the chyme as it is dehydrated and made into feces. The ileocecal sphincter regulates the amount of chyme entering the colon. The sphincter is usually tonically closed, but movement of chyme in the terminal ileum relaxes the sphincter, allowing small amounts of chyme to enter the colon. The **ileocecal sphincter** is controlled by the enteric nerves, as well as hormones (CCK and gastrin) and autonomic nerves.

Although segmental propulsion is very slow, **mass movements** periodically occur (usually 1 to 3 times daily). Mass movements are peristaltic in nature, but contractions extend over a longer period of time (seen in Fig. 22.8, descending colon). These strong contractions occur when the distal colon is relaxed and force the feces to travel quickly through the descending colon, into the rectum. The mass movements are stimulated by the parasympathetic nerves (vagal in the proxi-

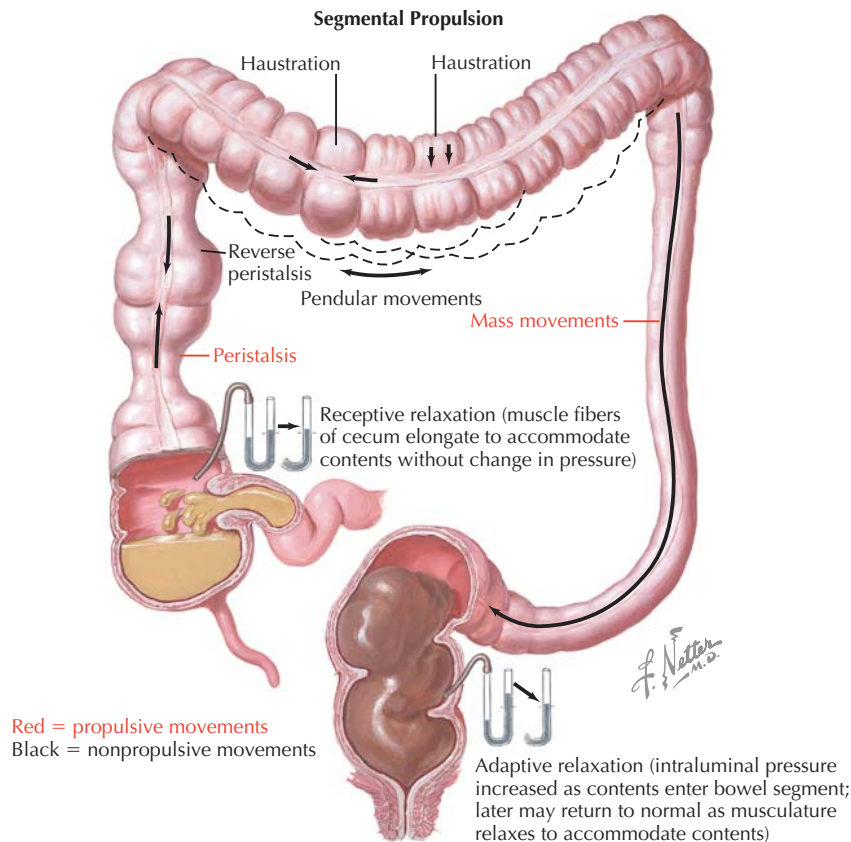


Figure 22.8 Motility in the Colon The colon is specialized for storage, and when bands of longitudinal muscle (taenia coli) contract, they form haustra. This is called segmental propulsion, which is very slow, and allows time for additional sodium and fluid absorption (colonic salvage). The slow movement dries out the chyme, producing feces. Several times a day mass movements will occur, creating peristaltic movements that force the chyme/feces toward the rectum. Factors that increase intestinal motility can cause diarrhea, by minimizing formation of haustrae and increasing mass movements.

mal colon, and pelvic nerves in the distal colon and rectum) as well as the hormones CCK and gastrin. These factors are all active when there is chyme in the upper GI tract (during active digestion), and their stimulation of mass movements serves to clear the lower GI tract of chyme and feces, in preparation for new waste coming down the tract. Conversely, stimulation of the sympathetic nerves (from the superior mesenteric in the proximal colon, and inferior mesenteric and superior hypogastric nerves in the distal colon) inhibits colonic movements, consistent with the overall actions of the sympathetic nervous system (SNS) on the GI system (Fig. 22.9).

DEFECATION AND THE RECTOSPINCTERIC (DEFECATION) REFLEX

When feces are moved into the rectum, the rectal distension is sensed by local mechanoreceptors that signal the myenteric nerves to relax the **internal anal sphincter** and initiate peristalsis, further pushing feces into the rectum. This local effect is reinforced by the parasympathetic nervous system (PNS),

which initiates stronger contractions via pelvic nerves. At the same time, an afferent signal is sent centrally, stimulating the **urge to defecate**. This is the **rectosphincteric, or defecation, reflex** (Fig. 22.10). This is a rapid reflex, and in response to the urge to defecate, we voluntarily constrict the **external anal sphincter**, to prevent defecation. If defecation cannot occur, there is some relaxation of the rectum, the internal anal sphincter constricts again, and the external anal sphincter can be relaxed, until another movement pushes feces into the rectum and the reflex recurs. At the appropriate time for defecation, the external anal sphincter voluntarily relaxes, intra-abdominal pressure increases (**Valsalva maneuver**), and defecation occurs. This reflex takes time to fully develop in children, and is usually intact by 2 to 4 years of age.

REGULATION OF MOTILITY

Motility is regulated by the enteric nerves, autonomic nerves, and hormones. The enteric nerves respond to luminal receptors that sense the presence and composition of the chyme, as

well as signals from autonomic nerves and hormones. As previously stated, the PNS primarily promotes motility, whereas the SNS slows or stops motility. A general rule is that most of the GI hormones inhibit gastric motility and/or emptying, thus allowing the duodenum to process the chyme that is released (Fig. 22.11). However, gastrin is an exception: it tends

to stimulate gastric motility and activity of the antral cycle, thus increasing gastric emptying. These hormones work together to regulate the exit of chyme from the stomach. The relaxation of the pylorus by gastrin and the PNS is moderated by the effects of gastric inhibitory peptide (GIP), CCK, and secretin, which increase pyloric tone.

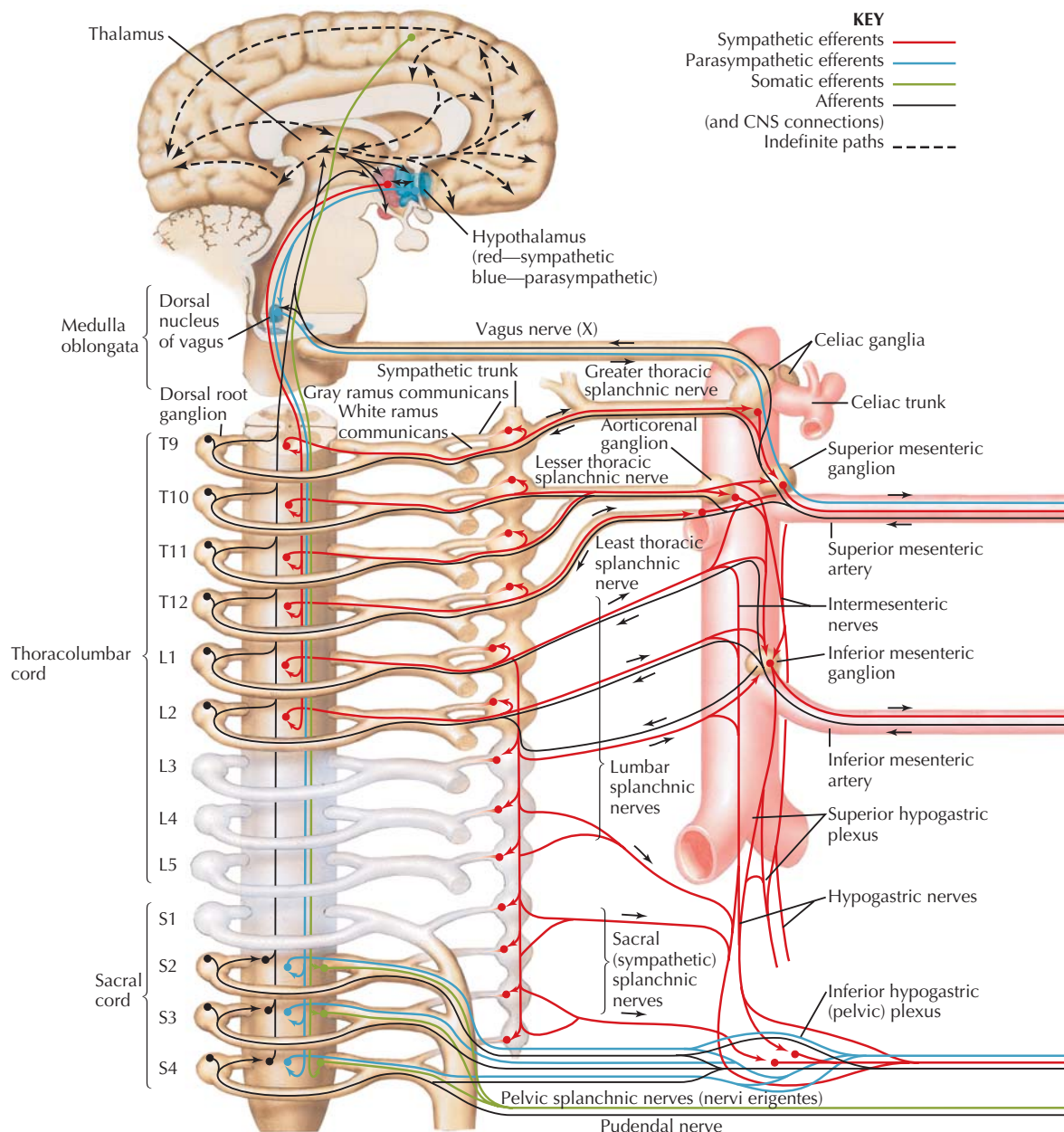


Figure 22.9 Autonomic Innervation of the Colon The small intestines and colon are innervated by both sympathetic and parasympathetic nerves. Sympathetic fibers from the spinal cord lead to the celiac, superior mesenteric ganglia, and inferior mesenteric ganglia. Parasympathetic fibers arise from the vagus (innervate through transverse colon) and pelvic (innervate descending colon, sigmoid, and rectum) nerves. Sympathetic stimulation reduces motility and secretions, whereas parasympathetic stimulation increases motility, secretions, and relaxation of the internal anal sphincter.

In addition, CCK and gastrin have stimulatory effects on motility in the lower small intestine and colon. They initiate mass movements in the colon, facilitating removal of the undigested waste matter. Motilin has the specific function of stimulating contractions during fasting (the MMC) (see Fig. 22.11).

VOMITING AND DIARRHEA

Vomiting is a reflex action that is under central regulation by the vomiting center in the medulla oblongata. Stimuli for vomiting also initiate salivation, and usually **retching** precedes vomiting. During retching, intestinal and gastric contents rise into the esophagus, but not into the mouth, due to a closed upper esophageal sphincter. The reflex can be stimulated by factors within the GI tract, as well as centrally through the chemoreceptor trigger zone (Box 22.1).

The events induced by all stimuli are consistent. **Reverse peristalsis** occurs (with aboral constriction, and relaxation behind the bolus) from the middle of the small intestine to the pylorus,

Box 22.1 Examples of Stimuli Initiating Vomiting Reflex

- Irritant in stomach or small intestine, enteric virus, or bacteria (local effect through enteric chemoreceptors)
- Systemic irritant sensed by chemoreceptor trigger zone in fourth ventricle of brain (near area postrema)
- Head injury (concussion) (central effect)
- Abnormal stimulation of vestibular organs (central effect via autonomic nervous system [ANS] to chemoreceptor trigger zone, then to vomiting center)

which relaxes to allow chyme to enter the stomach. The stomach fills, and strong abdominal muscle contractions force gastric contents into the esophagus, causing retching. With further stimulus, the upper esophageal sphincter relaxes, allowing contents to rise into the mouth. Because the vomitus includes intestinal and stomach contents, acid, bile, and electrolytes are excreted. Chronic vomiting can result in esophageal damage and metabolic disorders.

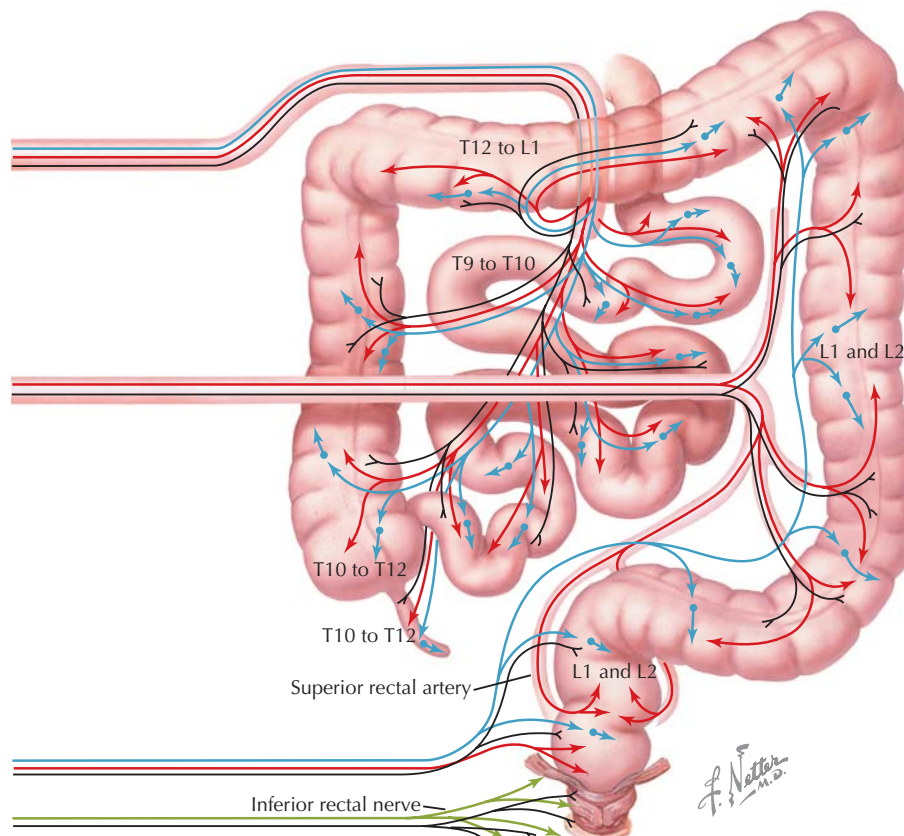


Figure 22.9 cont'd.

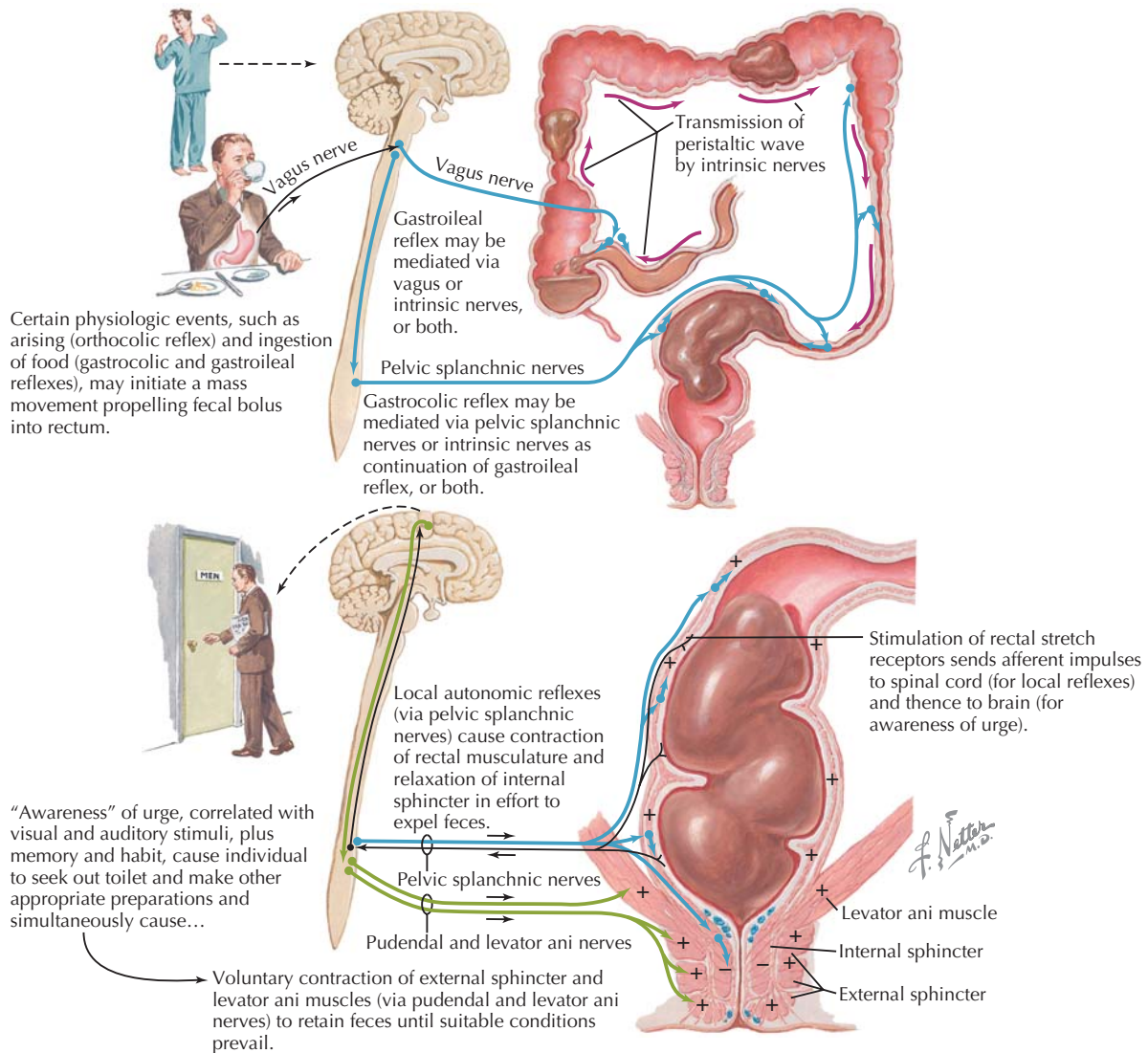


Figure 22.10 Defecation The sequence of events leading to defecation are detailed in this illustration. The defecation (or rectosphincteric) reflex is triggered by distension of the rectum by feces. This is sensed by the mechanoreceptors which activate the myenteric nerves, and impulses are transduced to the spinal cord (and back) to relax the internal anal sphincter (IAS). The spinal cord conveys signals to the brain to stimulate the urge to defecate. These reflex actions cause the voluntary contraction of the external anal sphincter (EAS) until defecation occurs. During defecation, the EAS relaxes, the person increases intra-abdominal pressure, and feces are eliminated from the rectum.

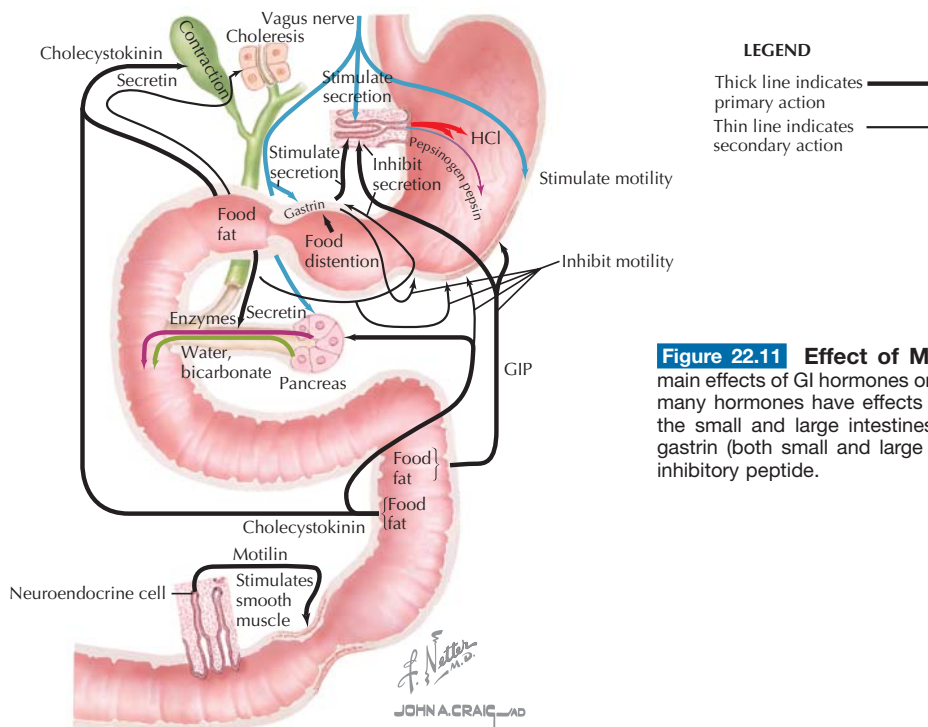
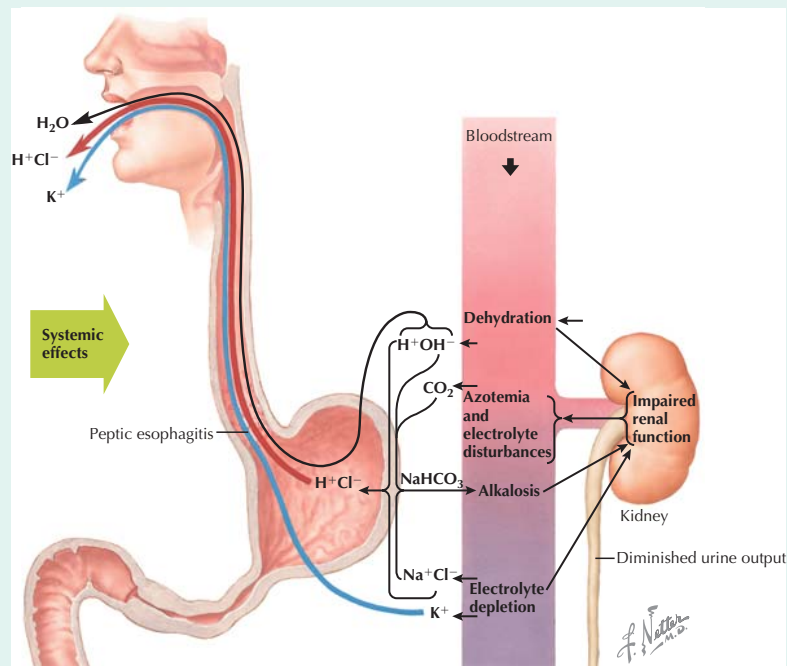


Figure 22.11 Effect of Major GI Hormones on GI Motility The main effects of GI hormones on GI motility are depicted in this scheme. While many hormones have effects on gastric motility, a few stimulate motility in the small and large intestines, including motilin (small intestine only) and gastrin (both small and large intestines, not shown on figure). GIP, gastric inhibitory peptide.



Think about homeostasis . . . With chronic vomiting or diarrhea, there is significant loss of fluids and electrolytes. Vomiting causes a disproportionate loss of H^+ and K^+ (from stomach acid). If vomiting continues over an extended time, in addition to dehydration, the loss of H^+ and K^+ can lead to metabolic alkalosis and hypokalemia (low plasma K^+).

In contrast, chronic diarrhea results in dehydration and a disproportionate loss of HCO_3^- , which can result in metabolic acidosis (see Chapter 20).



Vomiting Vomiting leads to a disproportionate loss of H^+ , Cl^- , and K^+ , as well as loss of fluid from the extracellular space. Prolonged vomiting results in dehydration, alkalosis, and hypokalemia (low plasma K^+).

Diarrhea occurs when chyme and fecal matter move too quickly through the colon. This can result from irritation of the lower GI tract (from bacteria or virus), diseases (irritable bowel syndrome, ulcerative colitis, Crohn's disease), or nervous stress. Movement in the colon is usually very slow (segmental propulsion), allowing absorption of almost all of the water and salt from the chyme, producing feces. If abnormal motility occurs, the fecal matter will move through the colon with minimal absorption, producing diarrhea. When this is in response to bacterial irritation, rapid motility serves to wash the irritant out of the tract, and the diarrhea abates with the loss of the bacteria or virus.

In general, **osmotic diarrhea** is the most common type of diarrhea. It is caused by hypermotility, which forces the

chyme/feces through the lower GI tract too rapidly for proper absorption of fluid and electrolytes. In contrast, **secretory diarrhea** is specifically caused by increased secretion of solutes into the lumen of the gut. The hallmark example of this occurs with **cholera toxin** (see "Cholera" Clinical Correlate).

REFLEXES

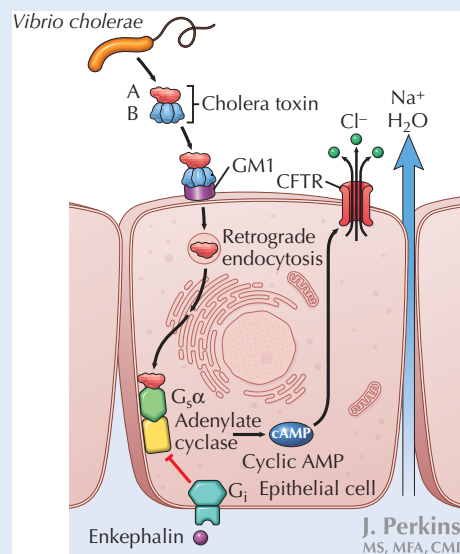
All of the reflexes ultimately serve to promote the efficient digestion and elimination of ingested material. One of the most important reflexes, the rectosphincteric reflex, was discussed earlier in relation to defecation. A variety of GI-related reflexes are described in Table 22.2.

CLINICAL CORRELATE

Cholera

Cholera is disease caused by the bacterium *Vibrio cholerae*, which is contracted in a fecal-oral manner, usually through ingestion of contaminated water. In the small intestine, the bacteria produce cholera toxin, which has devastating effects on the intestinal epithelium. The toxin modifies the G protein G_s , and as a result, GTPase activity is inhibited, causing prolonged activation of adenylyl cyclase. The resulting high level of cAMP activates the CFTR (see "Cystic Fibrosis" Clinical Correlate in Chapter 2), which causes active Cl^- secretion into the lumen of the small intestine. The Cl^- efflux is accompanied by Na^+ efflux (preserving electroneutrality), and water follows the electrolytes. This *isotonic*

secretion of electrolytes and water depletes the vascular space while maintaining plasma osmolarity. In this manner, the vascular space is rapidly contracted, causing severe dehydration and shock. Antibiotic therapy and intravenous hydration is the most desirable therapy; otherwise, oral rehydration therapy (ORT, containing water, electrolytes, and glucose) from an uncontaminated source can be given. The glucose in the ORT promotes sodium and electrolyte absorption and actually draws more of the secreted Na^+ and Cl^- back into the vascular space, along with water. This is effective in reducing the diarrhea and maintaining hydration. Without antibiotics, and with availability of uncontaminated water, the bacteria will be cleared from the GI tract in 7 to 10 days, as the enterocytes slough off and are replaced by new cells.



Actions of Cholera Toxin in Intestinal Cells

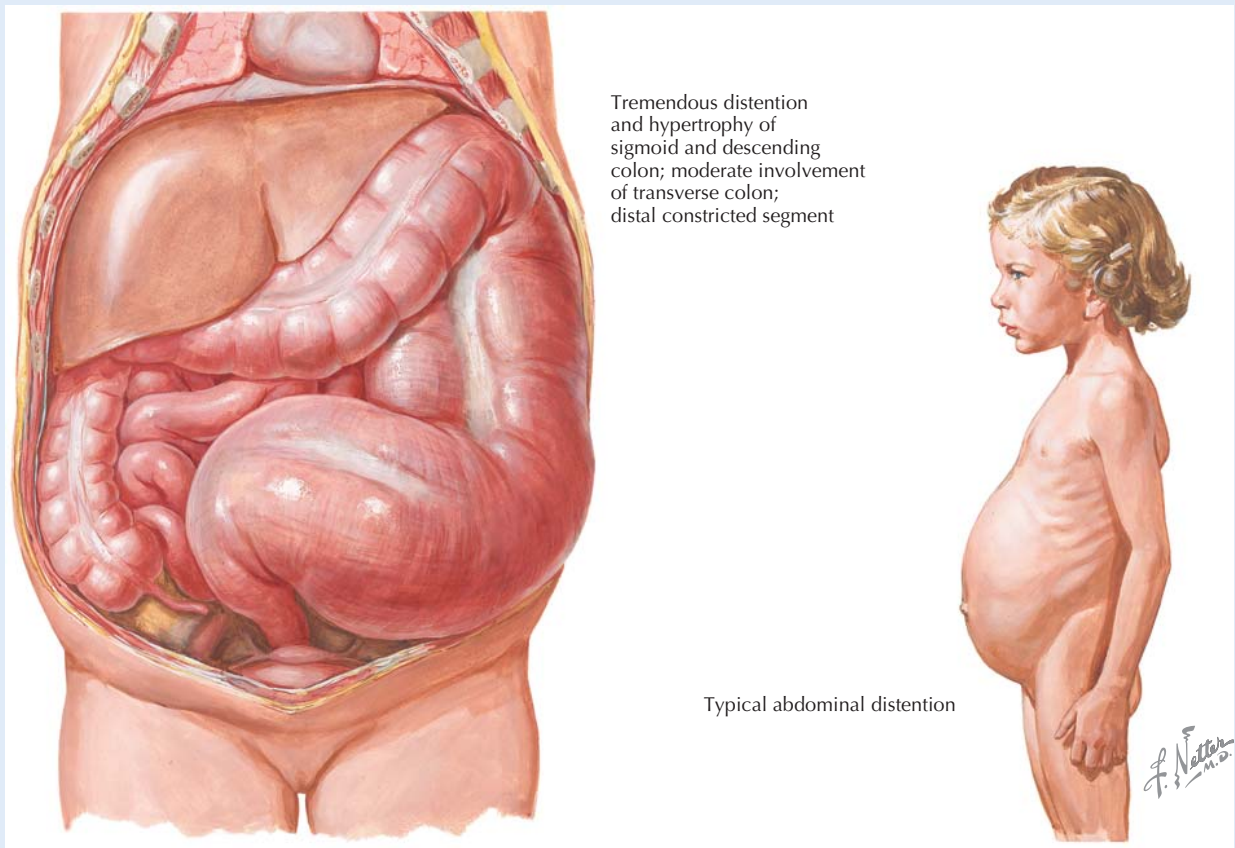
CLINICAL CORRELATE**Smooth Muscle Disorders: Achalasia and Hirschsprung's Disease**

The enteric nervous system (ENS) can be thought of as the central processing center responding to a variety of signals and making the appropriate connections. Because of this, if the ENS is absent from an area in the tract, pathology ensues. Smooth muscle diseases such as achalasia and Hirschsprung's disease involve aganglionic areas of the digestive tract (loss of nerves in some areas).

Achalasia is a loss of the ENS in the lower part of the esophagus and always involves the **lower esophageal sphincter**. The aganglionic region is not able to stimulate contractions and cannot transduce the signal to relax the lower esophageal sphincter. As a result, the sphincter tone is very high. Because the lower esophageal sphincter is not able to relax, the food and fluid can take hours, rather than seconds, to leave the esophagus. Over time (months to years) this causes dilation of the esophagus. This condition is very painful and can eventually lead to anorectic behavior. Also, because the ingested food sits in the esophagus for long periods of time, partial digestion occurs, and the contents can

further damage the esophageal mucosa. The etiology of achalasia is unknown, and treatment usually only partially alleviates the problem. Pharmacologic intervention with nitrites (isosorbide dinitrate) or calcium channel blockers (nifedipine or verapamil) is used but has limited efficacy, and the primary treatment is by balloon dilation or surgically cutting the sphincter (**esophagomyotomy**). This allows food and fluids to pass through the lower esophageal sphincter into the stomach but usually produces gastric reflux, and the patients must be treated with oral medication for GERD (gastroesophageal reflux disease).

Hirschsprung's disease (or **megacolon**) is a congenital disease that involves the loss of the ENS in the distal part of the colon and always involves the internal anal sphincter (IAS). The normal defecation reflex mediated by the myenteric plexus (feces in rectum → stretch → relaxation of IAS) does not occur, so very little defecation occurs and feces build up in the rectum and colon. This dilates the colon, and symptoms include little or no bowel movements and vomiting. Drugs are not effective; surgical removal of the aganglionic region typically restores the ability to defecate.



Hirschsprung's Disease (Megacolon)

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CHAPTER 23

Gastrointestinal Secretions

SALIVARY SECRETION AND REGULATION

Salivary Gland Secretion

Salivary secretions make foods easier to ingest (and digest) by lubricating, cooling, and adding digestive enzymes to the food. There are three sets of salivary glands producing the secretions: the parotid, the submandibular, and the sublingual. Secretion from these glands is under the control of cranial nerves VII and IX (Figs. 23.1 and 23.2 and Table 23.1). There are a few unique aspects of salivary gland secretion that bear noting:

- A large volume of fluid is produced by relatively small glands.
- Saliva is always hypotonic to plasma.
- Compared with plasma, saliva is relatively rich in K^+ .
- Salivary secretion is primarily regulated by cranial parasympathetic nerves, which control blood flow to the glands.

Approximately 1.5 liters (L) of saliva is secreted daily, most of which occurs with eating. When food enters the mouth, salivation is stimulated. The saliva serves to:

- Lubricate the food (water and mucins).
- Start digestion.
- Protect the mouth and gastrointestinal (GI) tract by cooling hot foods.
- Protect the tract through the antimicrobial actions of white blood cells (WBCs), IgA, and opsonins.
- Aid coagulation by secreting platelet-activating factor.
- Provide oral hygiene by buffering and cleansing the oral cavity—swallowing and spitting allow us to get rid of bacteria, whereas low flow while we sleep allows bacteria to accumulate, causing halitosis (bad breath).

Digestion of starch and lipids begins in the mouth. The serous glands secrete **salivary α -amylase** (or, **ptyalin**), which begins starch digestion by breaking α 1-4 linkages of glucose polymers (oligosaccharides). This is the same enzyme as pancreatic α -amylase. In addition, **Von Ebner's glands** of the tongue produce a serous secretion that includes the enzyme **lingual lipase**, which is active at low pH and can hydrolyze dietary triglycerides (TGs) in the stomach, yielding free fatty acids

(FA) and diglycerides. This is the same as gastric lipase but is different from pancreatic lipase (which hydrolyzes TGs to two FAs and one monoglyceride).

Production of Saliva

Salivary glands are exocrine glands, with secretions draining through ducts. The primary secretion is produced at the blind ends of the ducts, in the acini (Fig. 23.3). The glands are highly vascularized, and the parasympathetic nerves stimulate active secretion by increasing blood flow to the acini. Isotonic **ultrafiltrate** from the plasma diffuses through the acinar cells into the lumen, where this “**primary secretion**” is mixed with the additional products of the serous and mucous cells (α -amylase and mucins, respectively). The secretion drains down the striated ducts, and the electrolyte content is modified depending on the rate of flow through the ducts. Thus, as the rate of flow increases, there is less time for Na^+ and Cl^- reabsorption, and concentration of these electrolytes in saliva is higher (see Fig. 23.3B). Under conditions of reduced vascular volume (dehydration, hemorrhage), Na^+ and water reabsorption by the ductal cells is increased by **antidiuretic hormone (ADH)** and the mineralocorticoid **aldosterone** (see Chapter 19). The ADH increases water reabsorption (Na^+ follows), while aldosterone increases Na^+ reabsorption (water follows). Thus, when plasma volume is low, saliva production is decreased to conserve fluid. As indicated, regardless of flow rate, the saliva is always *hypotonic* to plasma.

When the saliva enters the mouth, it mixes with secretions from Von Ebner's glands (and other minor glands), adding lipase to the fluid. In addition, **R proteins** found in the saliva bind to the essential vitamin B12 in the stomach, protecting it from degradation. In the duodenum, pancreatic proteases cleave the R proteins and B12 binds to **intrinsic factor** (secreted by the gastric parietal cells), which protects it until it is absorbed in the terminal ileum. If any part of the pathway (especially intrinsic factor) is lost, B12 cannot be absorbed, and because B12 is critical for maturation of red blood cells, **pernicious anemia** develops.

Neural Control of Saliva

Salivary flow is dependent on parasympathetic nerves (glossopharyngeal and facial). They innervate the blood vessels of

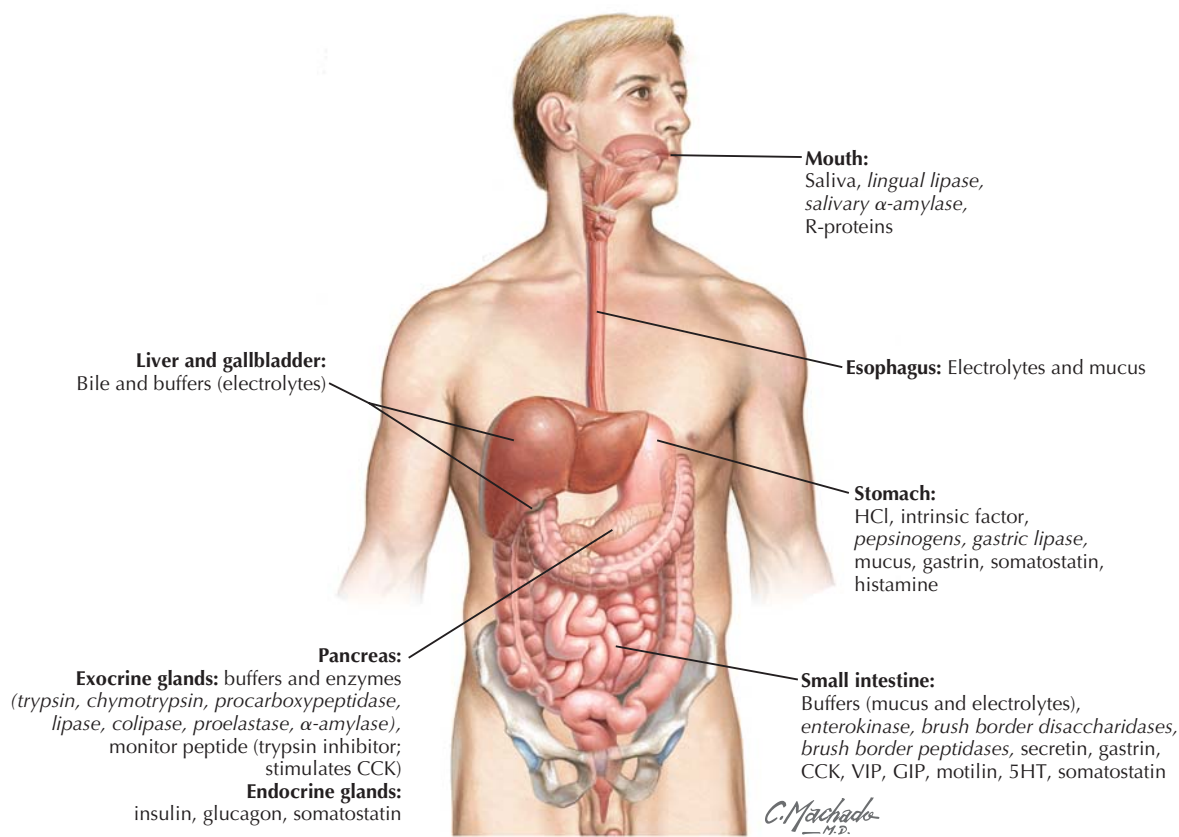


Figure 23.1 Major Secretions into the GI Tract In order to digest and absorb ingested nutrients, liters of secretions are added daily to the GI tract from associated glands, organs, and tissues. The general sources of digestive enzymes, acid, buffers, and bile are shown in the figure above. CCK, cholecystokinin; 5HT, serotonin; GIP, gastric inhibitory peptide; VIP, vasoactive intestinal peptide.

Table 23.1 Salivary Gland Secretions

Gland	Secretion	Cranial Nerve
Parotid	Watery fluid, without mucins (serous), ~25% of total	Glossopharyngeal (IX)
Submaxillary	Mixed serous/mucous fluid, ~70% of total	Facial (VII)
Sublingual	Mucous fluid, 5% of total	Facial (VII)

salivary glands, and when stimulated, increase blood flow, increasing ultrafiltrate secretion into the acini. If the parasympathetic nerves are severed, salivation will significantly decrease, and the glands will atrophy. The sympathetic nerves also innervate blood vessels around the glands as well as the myoepithelial cells in the acini. Sympathetic stimulation (for example, during stress) reduces blood flow but contracts the myoepithelial cells, releasing preformed mucous saliva. After

Table 23.2 Regulation of Salivary Flow

Increase Salivary Flow	Reduce Salivary Flow
• Parasympathetics (CN VII & IX), ACh, VIP • CNS (in cephalic, “sensory” phase) • Nausea • Esophageal distension • Chewy, flavorful foods • Dry, acidic foods • Meats, sweets, and bitter foods	• Sympathetics, NE • Hormones (ADH, aldosterone) conserve water and salt when volume depleted • Sleep • Dehydration (activates hormones) • Drugs, chemotherapy • Aging (decreases ANS tone; glands can atrophy)

ACh, acetylcholine; ADH, antidiuretic hormone; ANS, autonomic nervous system; CN, cranial nerve; CNS, central nervous system; NE, norepinephrine; VIP, vasoactive intestinal peptide.

that, flow will decrease (resulting in dry mouth). The salivatory nucleus in the brainstem transduces a sensory, central nervous system (CNS) effect, for example, when visual or olfactory input stimulates salivation (Table 23.2).

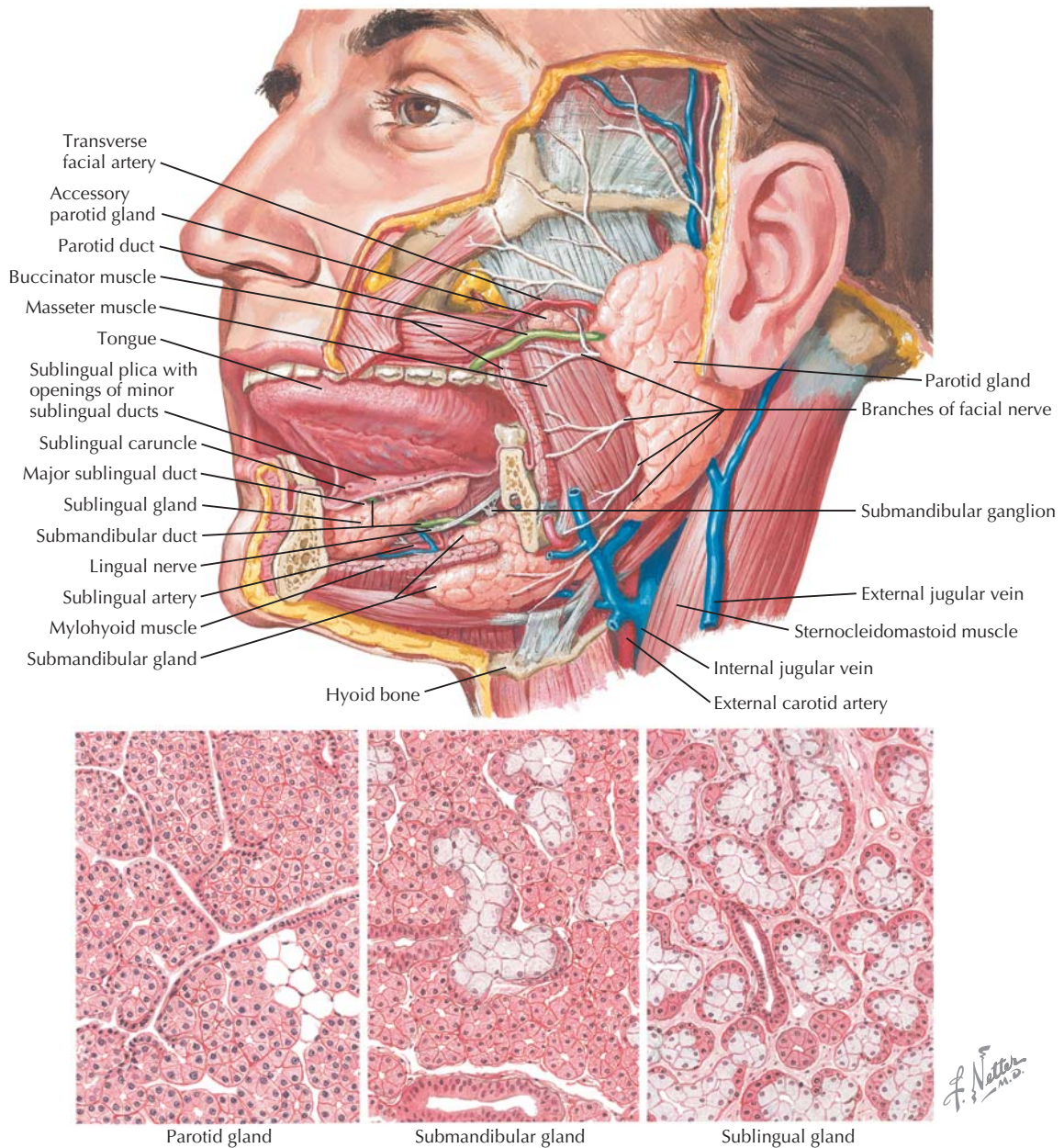


Figure 23.2 Salivary Gland Structure There are three paired salivary glands: the parotid, the submandibular, and the sublingual. The glands are specialized to secrete mainly serous (parotid), mucous (sublingual), or mixed (submandibular) saliva (see representative histology). About 1.5 liters of saliva is secreted daily, with the greatest secretion while eating.

Esophagus

Glands in the esophageal mucosa produce serous and mucous secretions, which lubricate the bolus of food and help propel it to the stomach. This is especially important if food gets stuck and secondary peristalsis occurs. The increased salivation and esophageal secretion aid movement of the food into the stomach.

GASTRIC PIT SECRETIONS AND REGULATION OF HCL SECRETION

The Gastric Pit

Gastric secretions facilitate and regulate digestion, lubricate the bolus of food, and protect the gastric mucosa. The mixing